

Optimizations

1. Proof of Concept #1:

Redid entire proof of concept.

Question 1

IV Characteristic for Resistors

0 / 2 pts

+ 2 pts Concept #1: I can create current voltage characteristics for a linear component and prove it with good documentation.

+ 1.5 pts Concept #1: I can create current voltage characteristics for a linear component, but my documentation needs improvement.

+ 0.5 pts Proof of concept well documented, but major technical errors are present.

+ 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

+ 0 pts Labels missing or inconsistent between schematic and results

+ 0 pts Results not legible or not presented in a logical order

✓ + 0 pts Completely missing or incorrect.

+ 0 pts Repeated submission. One submission per group!

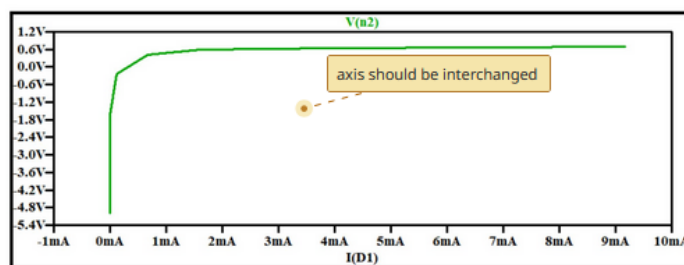
+ 0 pts Formatting and

Unfortunately, you don't do anything with the IV curve.

You effectively only show ohms law, that $V=IR$. The lab wanted you to look into the relation of the IV curve and ohms law.

2. Proof of Concept #2:

Changed the axis of the LT Spice simulation to have current on the y-axis and voltage on the x-axis. Rewrote discussion.



is hard to truly identify the issue. Will revisit once fixed. Without Waveforms too I cannot start part 3 without this section being likely incorrect.

Include more detailed discussion with a good analysis

Question 2

IV Characteristic for Diode

1.5 / 2 pts

+ 2 pts Concept #2: I can create current voltage characteristics for a non-linear component and prove it with good documentation.

✓ + 1.5 pts Concept #2: I can create current voltage characteristics for a non-linear component, but my documentation needs improvement.

+ 0.5 pts Proof of concept well documented, but major technical errors are present.

+ 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

+ 0 pts Labels missing or inconsistent between schematic and results

+ 0 pts Results not legible or not presented in a logical order

+ 0 pts Completely missing or incorrect.

+ 0 pts Repeated submission. One submission per group!

3. Proof of Concept #3:

Added in discussion.

Discussion (and answer related questions in Alpha Lab):

Comparison of Analysis, Simulation and Measurement results. Both a simple summary of results (like a numerical chart of values) and a simple description that details if the results are as you expect. Also include any speculation as to why they may be different from one another if they are different. What variation is too much for example...explore this.

Include a table that compares mathematical, simulation, and experimental results

where is the discussion

Question 3

Differential Resistance

1.5 / 2 pts

+ 2 pts Concept #3: I can demonstrate that the differential resistance varies along the IV curve of a non-linear component and I can prove it with good documentation.

✓ + 1.5 pts Concept #3: I can demonstrate that the differential resistance varies along the IV curve of a non-linear component, but my documentation needs improvement.

+ 0.5 pts Proof of concept well documented, but major technical errors are present.

+ 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

+ 0 pts Labels missing or inconsistent between schematic and results

+ 0 pts Results not legible or not presented in a logical order

+ 0 pts Completely missing or incorrect.

+ 0 pts Repeated submission. One submission per group!

Proof of Concepts Lab 2 Omega

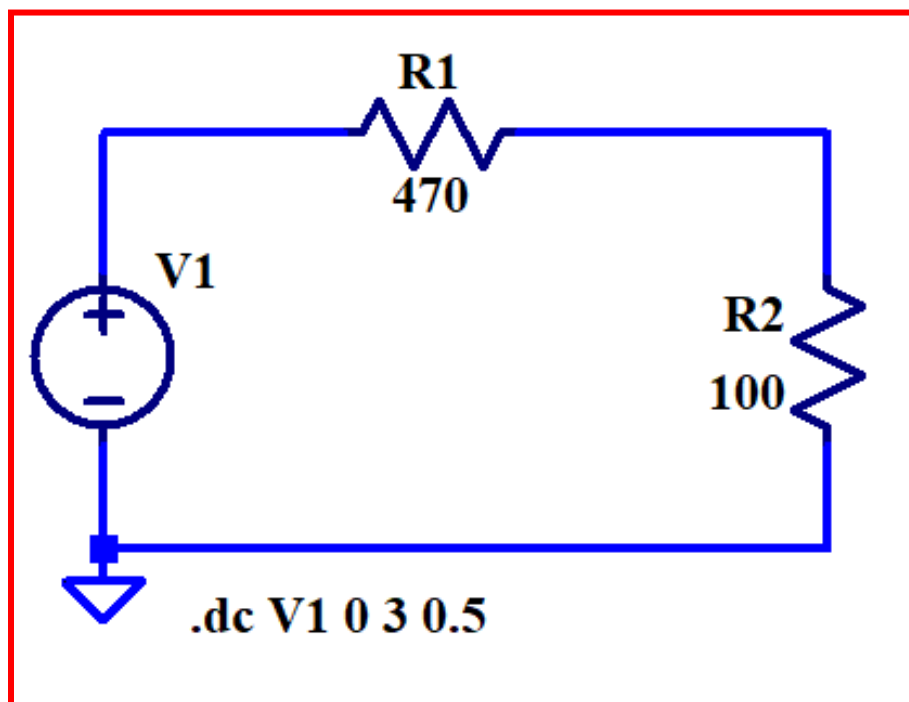
Jacob: Part A

Kimberly: Part B

Evan: Part C

1. (A) Prove that the slope of an I-V curve corresponds with ohm's law for two different resistor values

Building Block:



Description:

To show the IV characteristic of a linear component, analyze a resistor in a circuit with a DC power supply. Since the IV characteristic of a resistor is linear, then the plotted line of current on the y axis and voltage on the x axis should have a slope equal to 1 divided by the resistors value.

Analysis:

Ohm's law allows us to analyze the I and V of a linear component, a resistor.

Current (I) - The flow of electric charge through the resistor.

Voltage (V) - The potential difference across the resistor.

Resistance (R) - The value of the resistor, either R1 or R2.

Rewriting Ohm's Law:

$$V = IR \Rightarrow I = \frac{V}{R} \Rightarrow I = V \cdot \frac{1}{R}$$

The formula follows a similar form to $y = mx + b$ where m is the slope in this case when plotting I, current on the y axis and V, voltage on the x axis demonstrating the IV characteristic of a linear component which in this case is a resistor. Therefore, the slope of the IV curve of a resistor should be equal to $1/R$, R being the resistance of the resistor being measured..

The slope of the IV curve for R1 should be $\frac{1}{470} \approx 0.00213$ and R2 should be $\frac{1}{100} = 0.01$.

To prove this, analyze the voltage for R1 and R2 with different values of V1 and make a plot for when

V1 = 0, 0.5, 1, 1.5, 2, 2.5, and 3V

To get current, analyze as a single resistor circuit:

$$R_{Total} = R_1 + R_2 = 470 + 100 = 570\Omega$$

And so,

$$I = \frac{V_1}{R_{Total}}$$

And for voltage of a resistor where n is the resistor number,

$$V_{Rn} = IR_n$$

When $V_1 = 0V$:

$$I = \frac{0}{570} = 0 A$$

$$V_{R1} = 0 * 470 = 0V$$

$$V_{R2} = 0 * 100 = 0V$$

When $V_1 = 2V$:

$$I = \frac{2}{570} = 0.0035088 A$$

$$V_{R1} = 0.0035088 * 470 = 1.6491V$$

$$V_{R2} = 0.0035088 * 100 = 0.3509V$$

When $V_1 = 0.5V$:

$$I = \frac{0.5}{570} = 0.0008772 A$$

$$V_{R1} = 0.0008772 * 470 = 0.4123V$$

$$V_{R2} = 0.0008772 * 100 = 0.0877V$$

When $V_1 = 2.5V$:

$$I = \frac{2.5}{570} = 0.0043860 A$$

$$V_{R1} = 0.0043860 * 470 = 2.0614V$$

$$V_{R2} = 0.0043860 * 100 = 0.4386V$$

When $V_1 = 1V$:

$$I = \frac{1}{570} = 0.0017544 A$$

$$V_{R1} = 0.0017544 * 470 = 0.8246V$$

$$V_{R2} = 0.0017544 * 100 = 0.1754V$$

When $V_1 = 3V$:

$$I = \frac{3}{570} = 0.0052632 A$$

$$V_{R1} = 0.0052632 * 470 = 2.4737V$$

$$V_{R2} = 0.0052632 * 100 = 0.5263V$$

When $V_1 = 1.5V$:

$$I = \frac{1.5}{570} = 0.0026316 A$$

$$V_{R1} = 0.0026316 * 470 = 1.2369V$$

$$V_{R2} = 0.0026316 * 100 = 0.2632V$$

For R1:

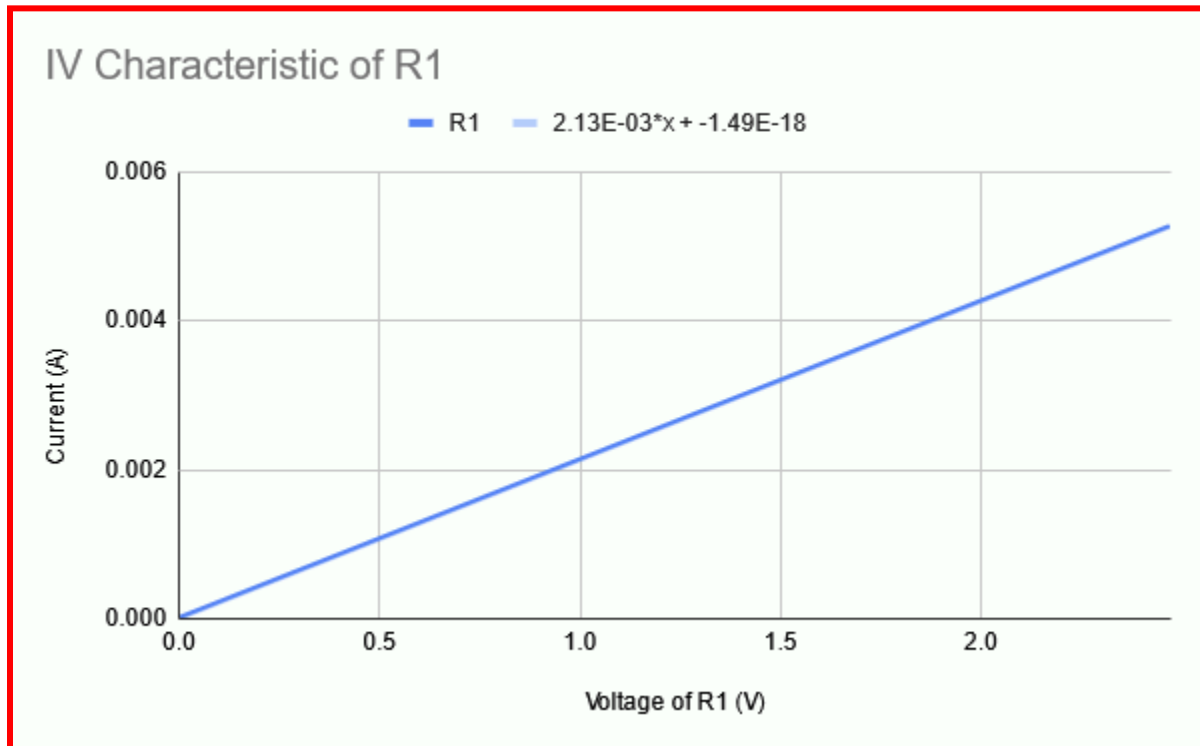
Voltage (V)	Voltage of R1 (V)	Current (mA)
0	0	0
0.5	0.4123	0.8772
1	0.8246	1.7544
1.5	1.2369	2.6316
2	1.6491	3.5088
2.5	2.0614	4.386
3	2.4737	5.2632

For R2:

Voltage (V)	Voltage of R2 (V)	Current (mA)
0	0	0
0.5	0.0877	0.8772
1	0.1754	1.7544
1.5	0.2632	2.6316
2	0.3509	3.5088
2.5	0.4386	4.386
3	0.5263	5.2632

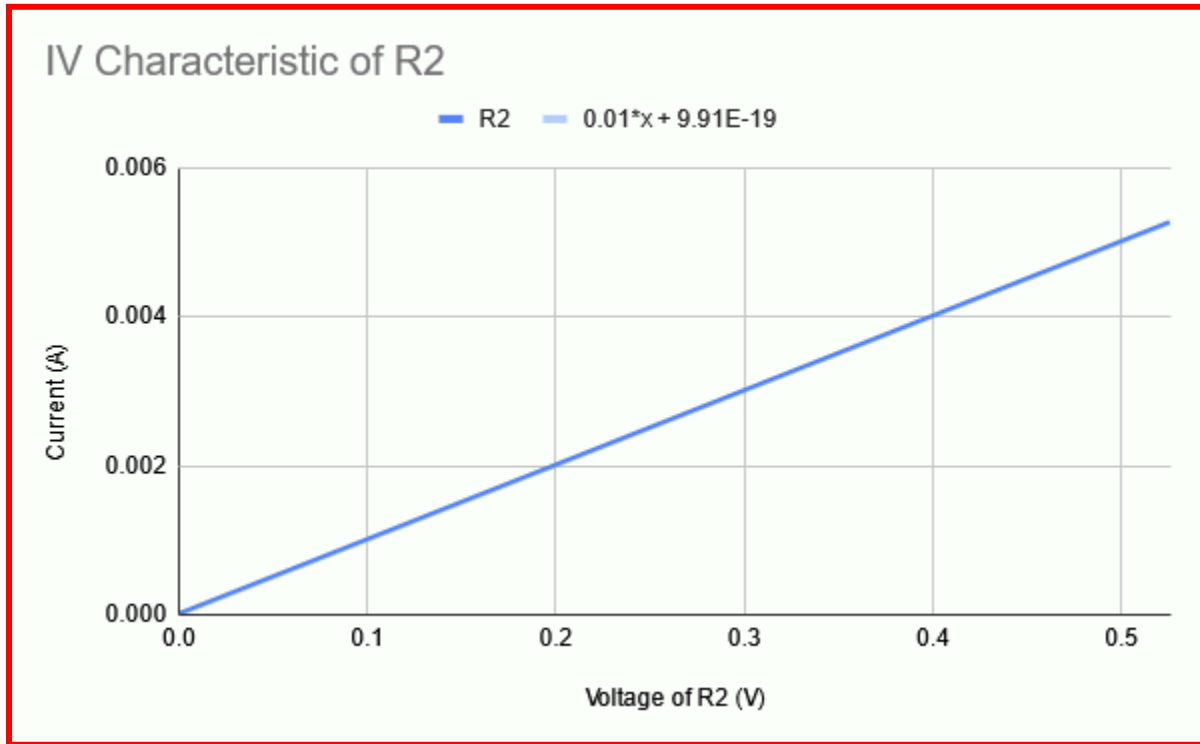
When plotting the data in sheets, to have voltage on the x axis and current on the y axis, only one resistor's data can be plotted on a graph.

For R1:



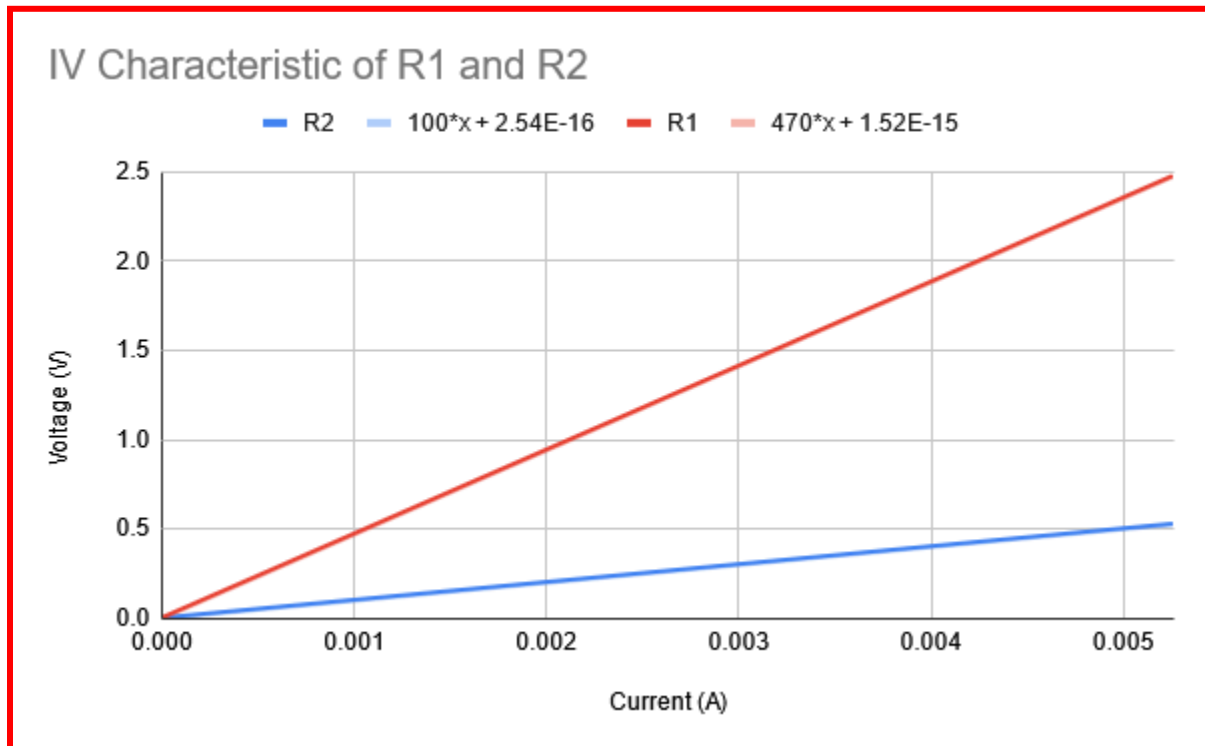
The slope of the trendline for R1 with voltage on the x axis and current on the y axis results in a slope of 0.00213 which is equal to our expected slope for R1. To confirm the value of R1, do $1/0.00213 = 469.484$ which is around the value of $R1 = 470\Omega$.

For R2:



Plotting the IV curve of R2 with voltage on the x axis and current on the y axis, the resulting trendline has a slope of 0.01 which is equal to our expected slope of 0.01. To confirm the value of R2, do $1/0.01 = 100$ which also matches our value of $R2 = 100\Omega$.

And on a combined graph with R1 and R2 but with current on the x axis and voltage on the y axis:

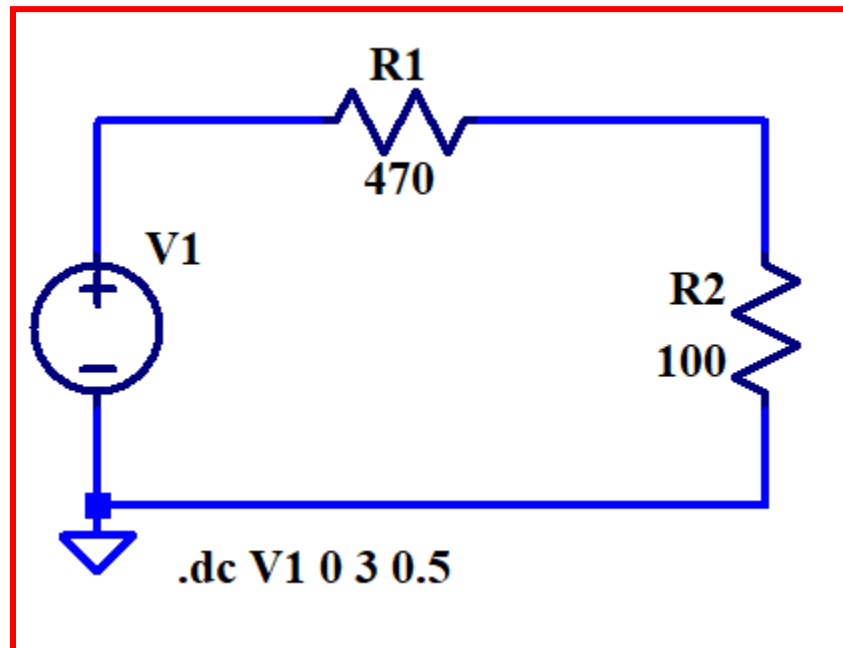


The slope of the trendline of R1 and R2 on a graph with current on the x axis and voltage on the y axis is equal to the value of each respective resistance. This graph follows Ohm's law formatted as $V = IR$ where R is m, V is y, and I is x.

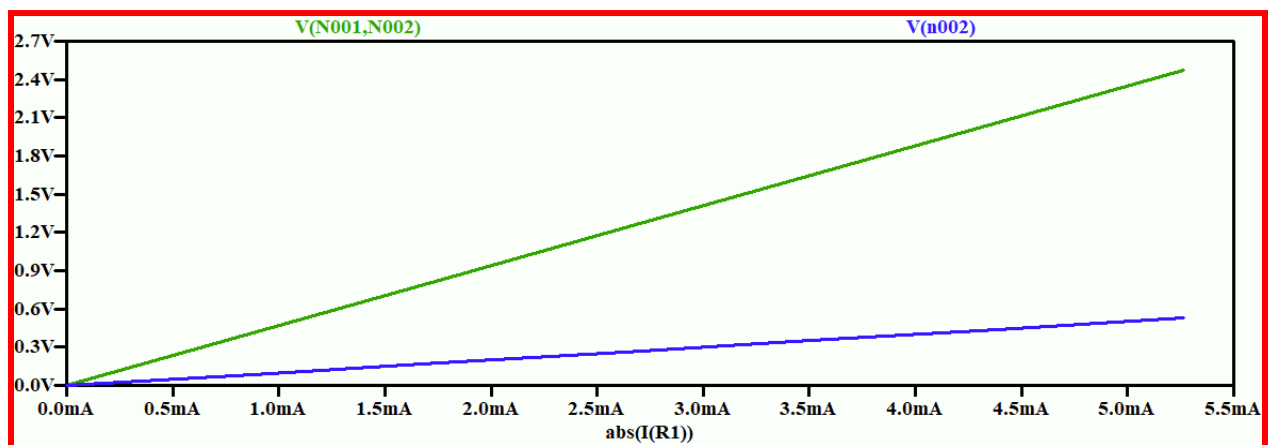
Looking at the trendline for R1, the slope is 470 so the value of R1 should be 470Ω.

And for R2, the slope is 100 so the value of R2 should be 100Ω.

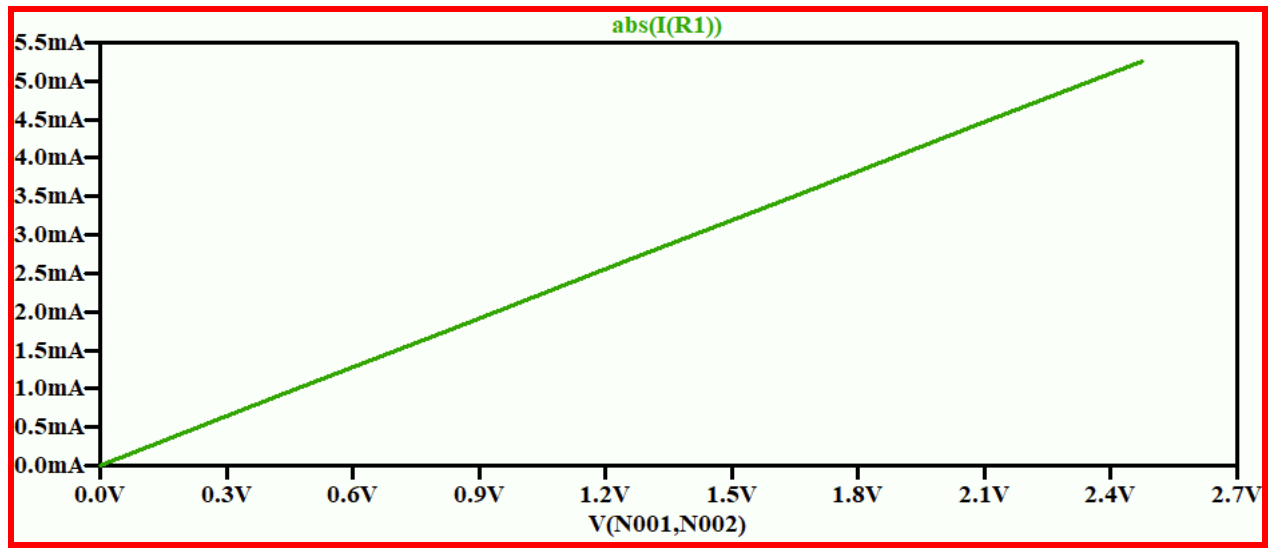
Simulation:



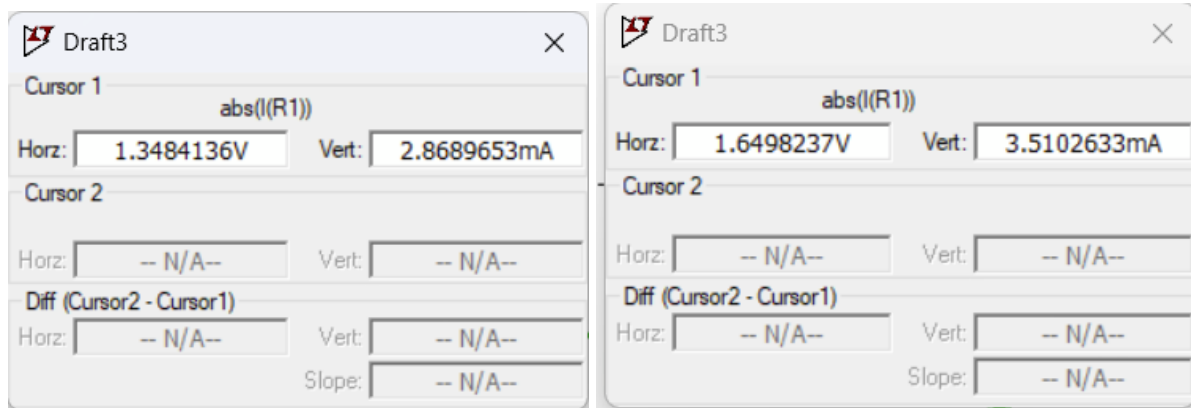
I can only plot the IV characteristic for each resistor one at a time when voltage is on the x axis and y axis so to have both on one plot, have voltage on the y axis and current on the x axis just to demonstrate how the IV characteristic for each resistor has a different value for a slope as the values of resistance for R1 and R2 are also different.



For R1:



Where $V(N001, N002)$ is the x-axis and is the voltage at R1 and current is on the y-axis. The slope of the IV curve should be $1/R1$. To calculate the slope, take 2 points using active curve trace and find the slope between the 2 points.

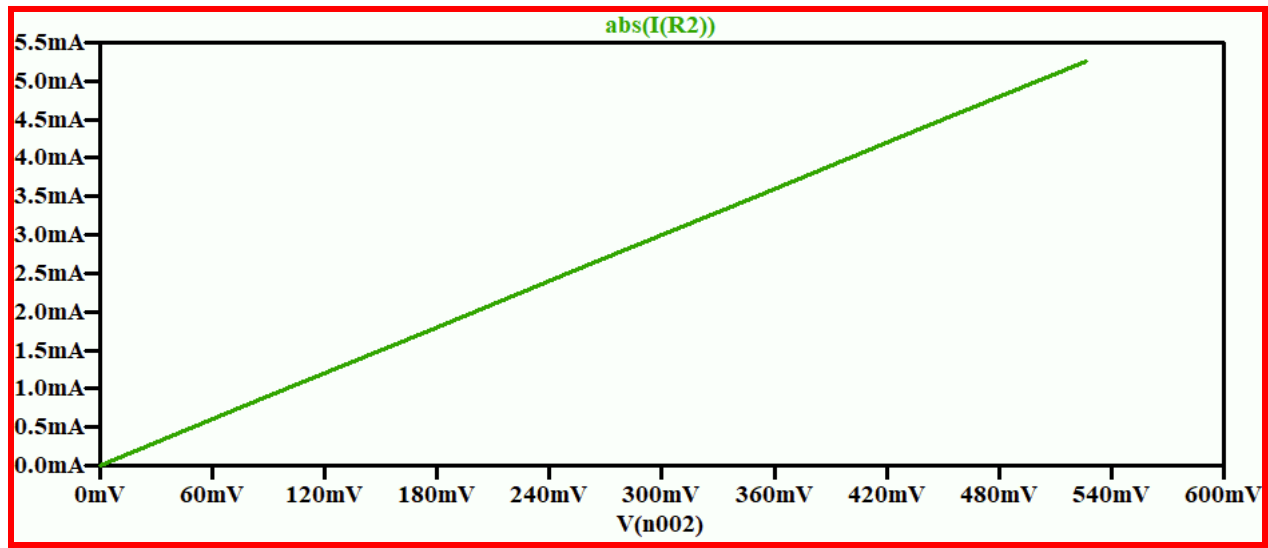


Using points (1.3484136, 0.0028689653) and (1.6498237, 0.0035102633), use the slope formula:

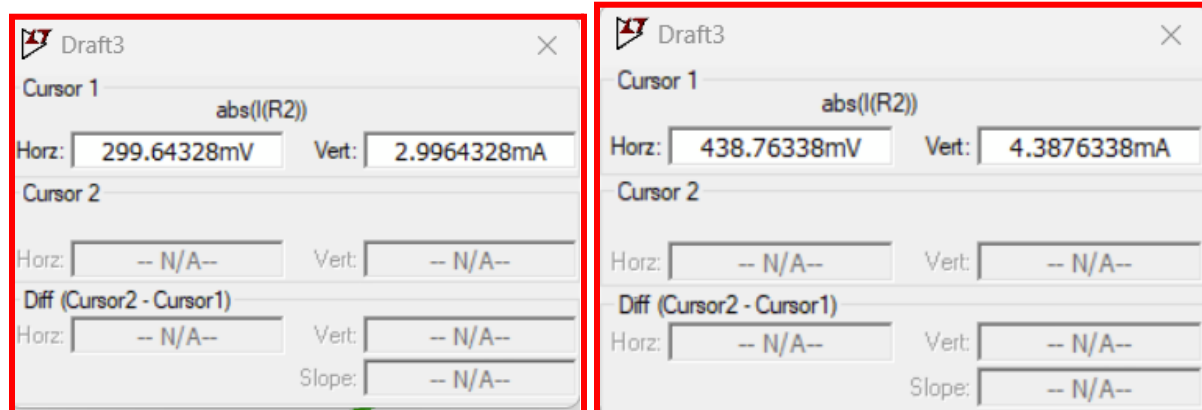
$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{0.0035102633 - 0.0028689653}{1.6498237 - 1.3484136} = \frac{0.000641298}{0.3014101} = 0.00212765929$$

The slope is around the value calculated before ≈ 0.00213 and to check for the value of R1, since the slope is $1/R1$, do $1/0.00212765929 \approx 470\Omega$

For R2:



With V(n002), the voltage at R2, on the x-axis and current on the y axis, the slope of the IV curve for R2 should be $1/R2$. To check the slope, use an active trace on the line to pull 2 points and find the slope between them.



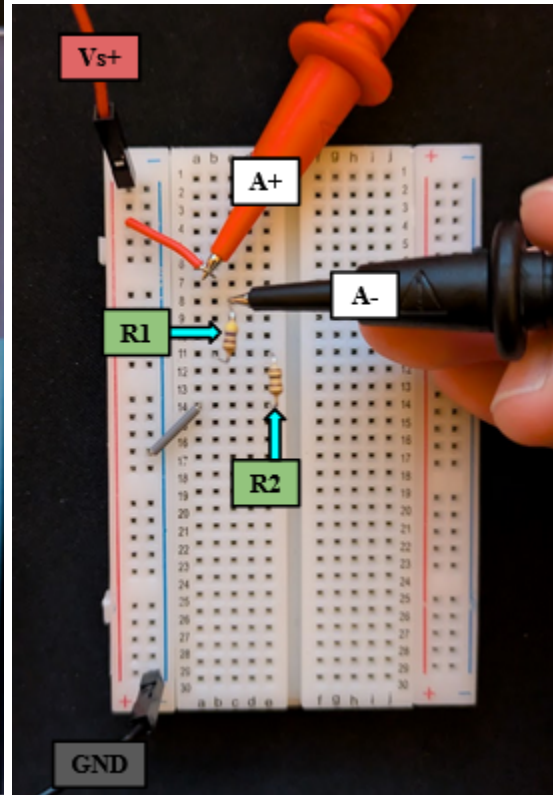
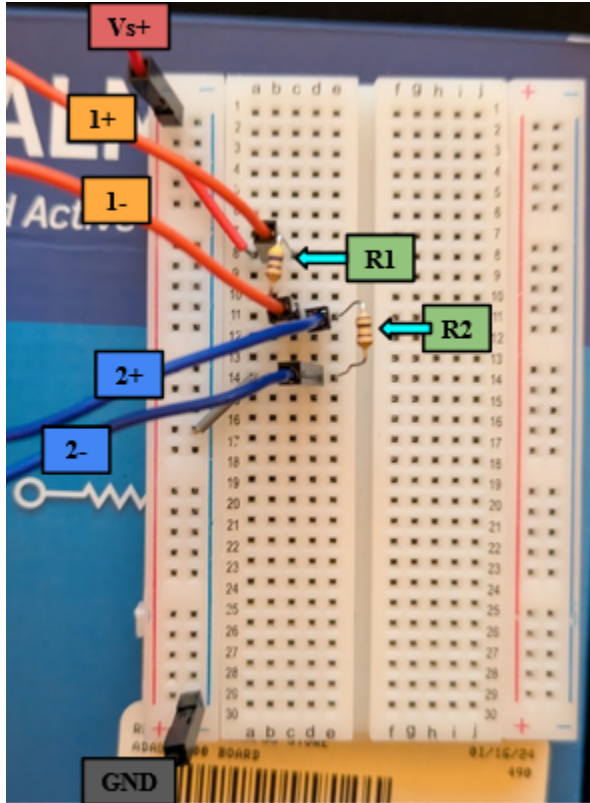
Use points (0.29964328, 0.0029964328) and (0.43876338, 0.0043876338) to calculate slope.

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{0.0043876338 - 0.0029964328}{0.43876338 - 0.29964328} = \frac{0.001391201}{0.1391201} = 0.01$$

The slope is calculated to be 0.01 which matches our analysis from above. To confirm the value of R2, since the slope = $1/R2$, do $1/0.01 = R2 = 100\Omega$

Measurement:

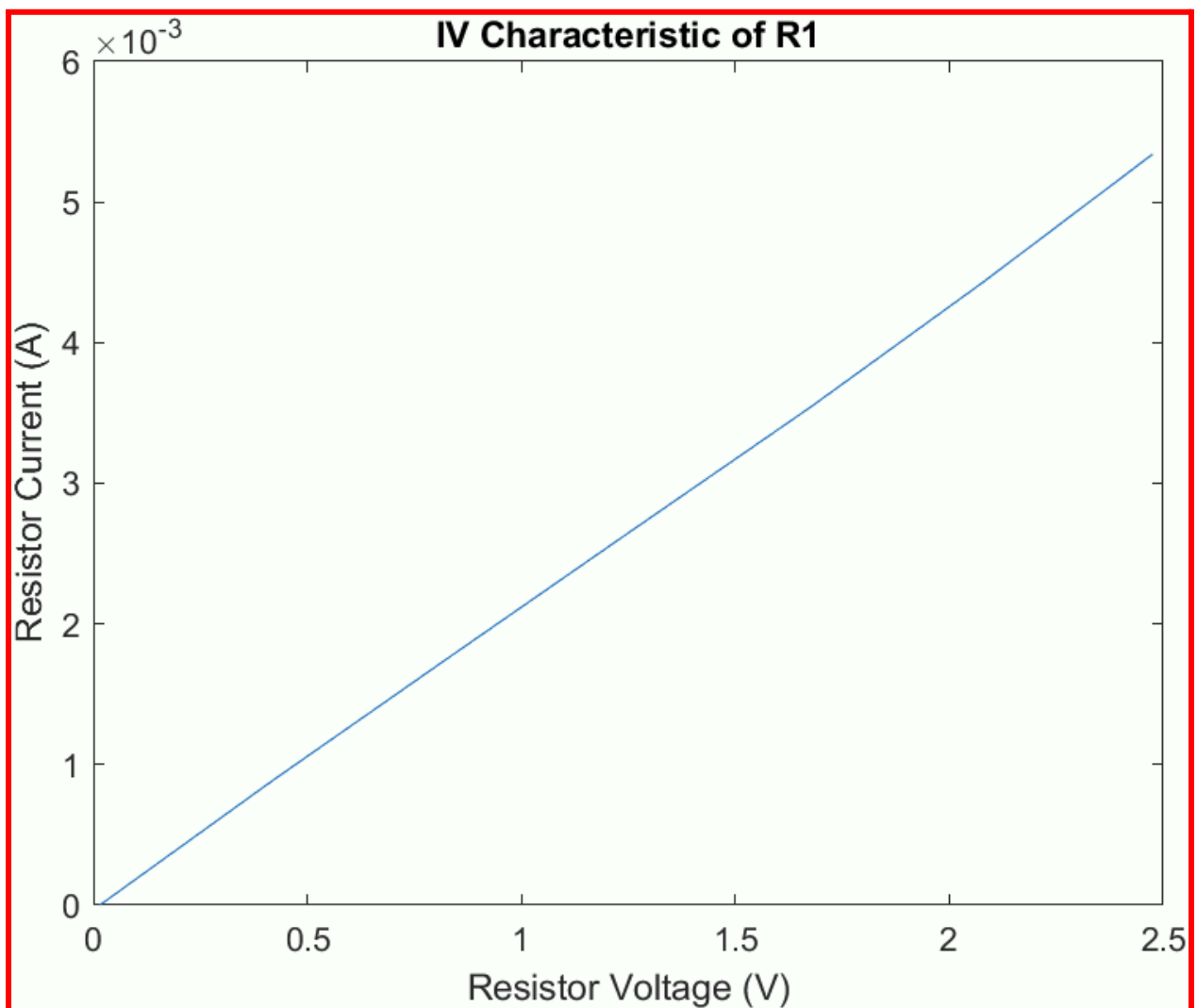
On scope measure $V(R1)$ on channel 1 and $V(R2)$ on channel 2 and use a multimeter to measure current for each value of V_s .



Voltage (V)	R1 Voltage (V)	R2 Voltage (V)	Current (mA)
0	0.015	0	0
0.5	0.416	0.087	0.88
1	0.830	0.175	1.76
1.5	1.242	0.265	2.63
2	1.675	0.354	3.54
2.5	2.079	0.442	4.43
3	2.475	0.531	5.34

For R1:

```
lab2part1.m  x  +
1  I = [0 0.00088 0.00176 0.00263 0.00354 0.00443 0.00534];
2  V = [0.015 0.416 0.83 1.242 1.675 2.079 2.475];
3  coeffs = polyfit(V, I, 1);
4  plot(V, I)
5  xlabel('Resistor Voltage (V)')
6  ylabel('Resistor Current (A)')
7  title('IV Characteristic of R1')
8
```





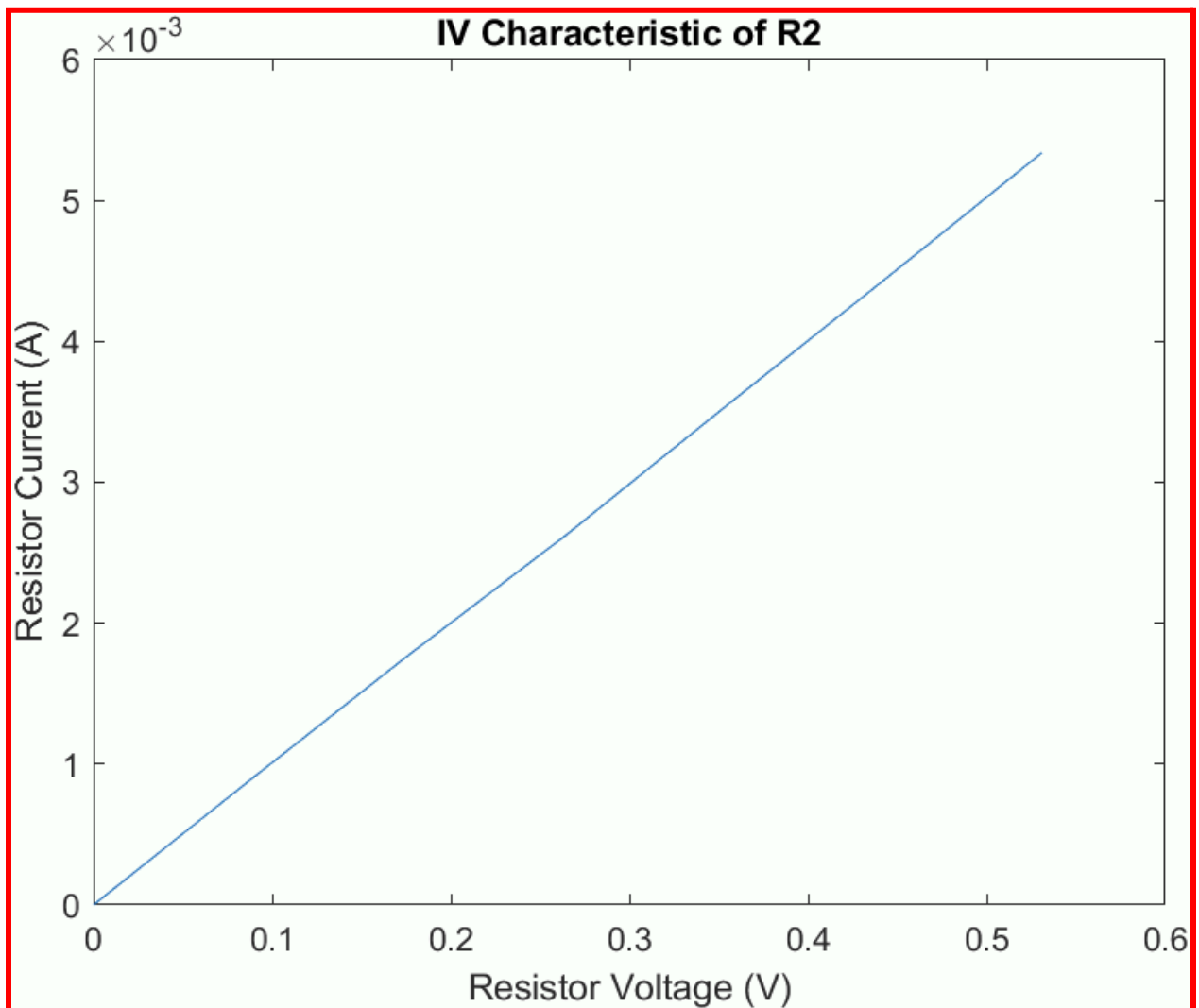
coeffs	
1x2 double	
	1
1	0.0022

MATLAB finds the linear regression slope to be 0.0022. This value closely matches our resulting slope value from analysis of 0.00213. Since this slope should equal to $1/R$, to get our supposed value of R , do $1/0.0022 = 454.55$.

Similarly, by measuring with a multimeter, the resistance of $R1$ is around 462Ω which is similar to our value from MATLAB

For R2:

```
lab2part1.m  x  +
1      I = [0 0.00088 0.00176 0.00263 0.00354 0.00443 0.00534];
2      V = [0 0.087 0.175 0.265 0.354 0.442 0.531];
3      coeffs = polyfit(V, I, 1);
4      plot(V, I)
5      xlabel('Resistor Voltage (V)')
6      ylabel('Resistor Current (A)')
7      title('IV Characteristic of R2|')
8
```





coeffs		
1x2 double		
	1	
1	0.0100	-2

From MATLAB, the slope of the linear regression line is 0.01 which is equal to $1/R$. This value also matches our slope value from analysis which was 0.01. To verify R , do $1/0.01 = 100$ meaning the value of R_2 should be around 100Ω .

When measuring the resistance of R_2 with a multimeter, it reads to be around 101Ω which is similar to the MATLAB value.

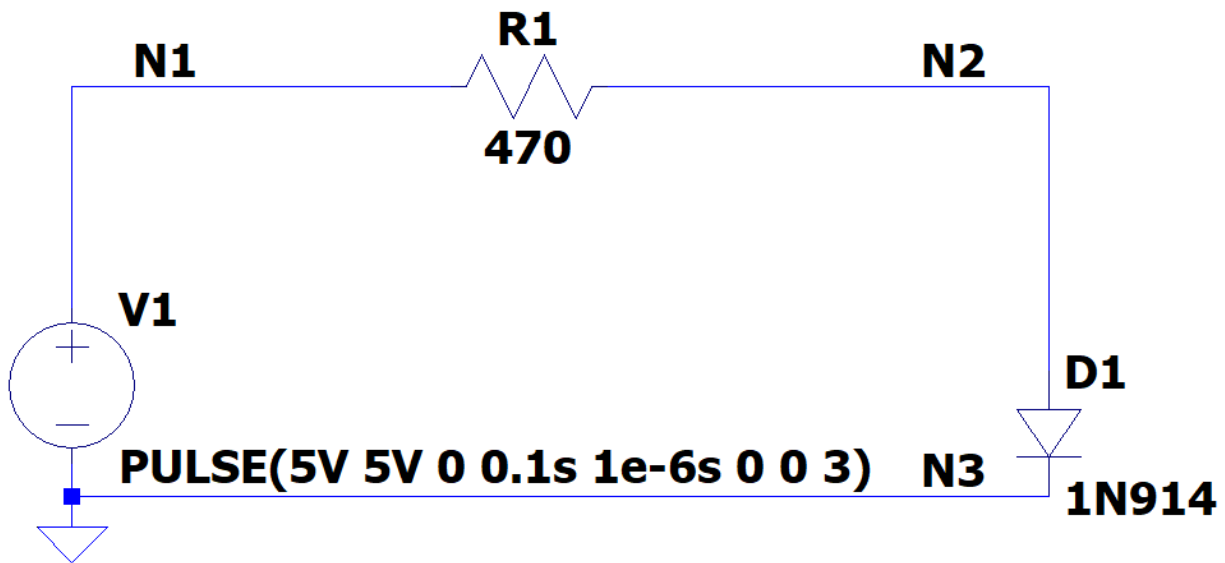
Discussion (and answer related questions in Alpha Lab):

Resistance	Analysis	Simulation	Measurement	% Err
R1 (470Ω)	470	470	454.55	3.29%
R2 (100Ω)	100	100	100	0

The results from the mathematical analysis, simulation and measurement match up to prove that the IV curve of the resistors have a slope of equal value to the resistor itself. This follows the concept of Ohm's law that demonstrates a linear relationship between voltage, current and resistance where $V = IR$ can be rewritten to demonstrate the IV curve with I as the y-axis and V as the x-axis as $I = V/R$. This is a similar model of $y = mx + b$ which is the formula for a linear line. The slope of the resulting line should then be $1/R$ as the value for m of the line.

2. (A) Prove the non-linear I-V curve for a light emitting diode

Building Block:



Description:

In this demonstration, we will show that the current-voltage (I-V) curve of a light-emitting diode (LED) is non-linear. This non-linearity comes from the semiconductor properties of the diode, creating an exponential relationship between current and voltage. We will look at the I-V characteristics of the LED by changing the voltage and measuring the current.

Analysis:

Diode Eqn

$$I = I_s(e^{(V/nV_t)} - 1)$$

KVL Eqn:

$$V_s = V_r + V_{led}$$

Thermal Voltage:

$$V_t = kT/q$$

KVL plugged in

$$5V = I \cdot 470\Omega + V_d$$

Thermal Voltage plugged in

$$V_t \approx 26\text{mV (assumed room temp)}$$

Diode Eqn

$$I = 10^{-7}(e^{0.0265V} - 1) \approx 10^{-7}(e^{192.3V} - 1)$$

I is the current through the diode.

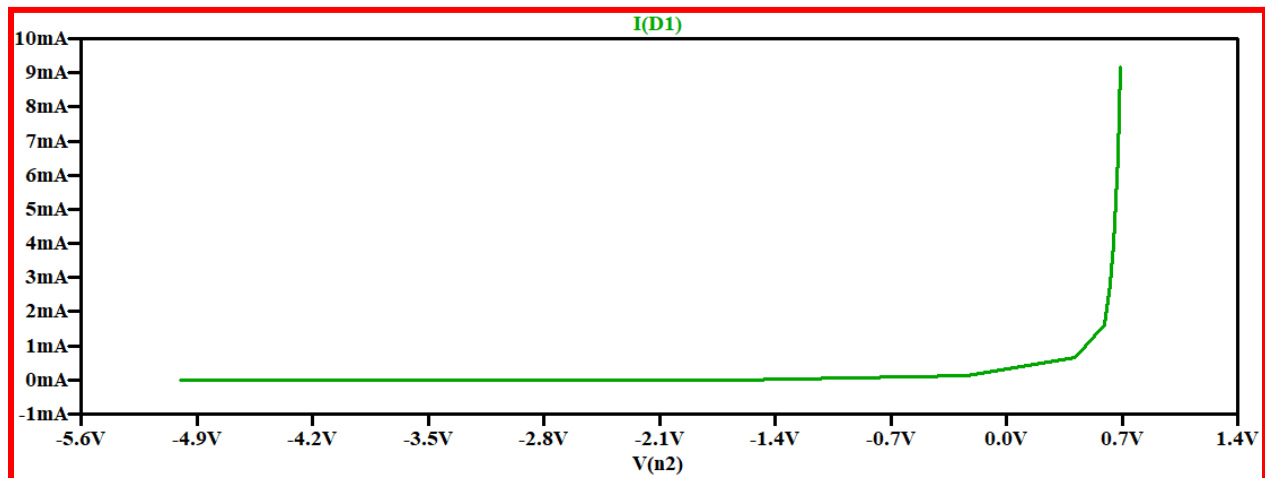
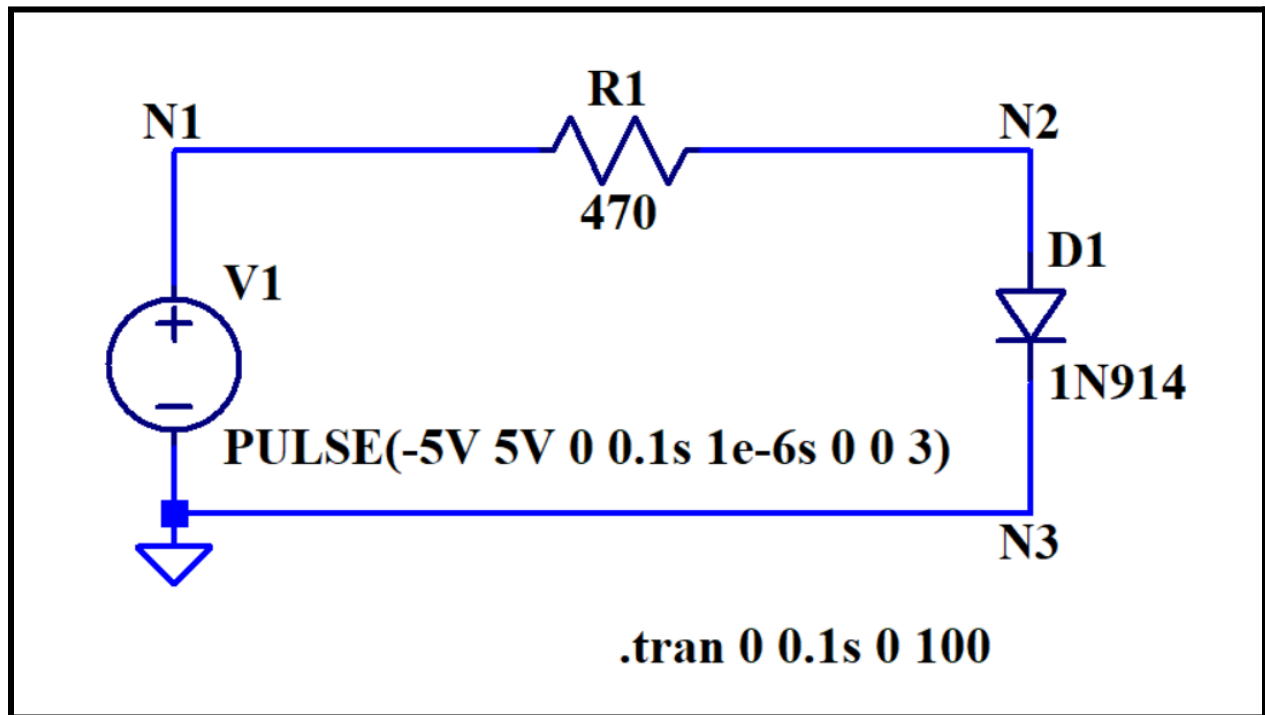
I_s is the reverse saturation current (a small constant).

V is the voltage across the diode.

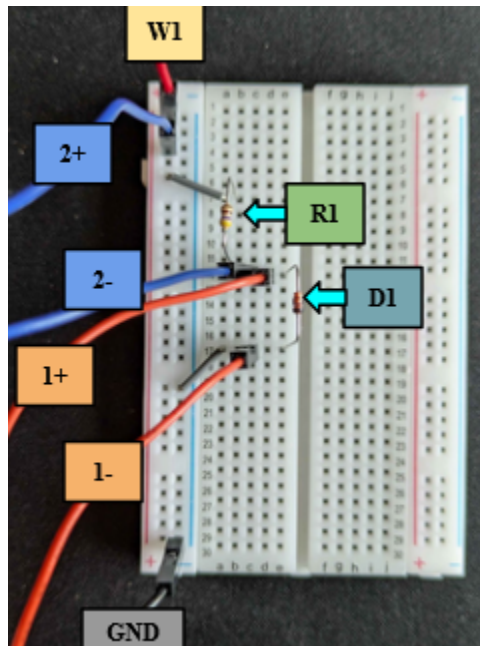
n is the ideality factor (typically between 1 and 2).

V_t is the thermal voltage

Simulation:



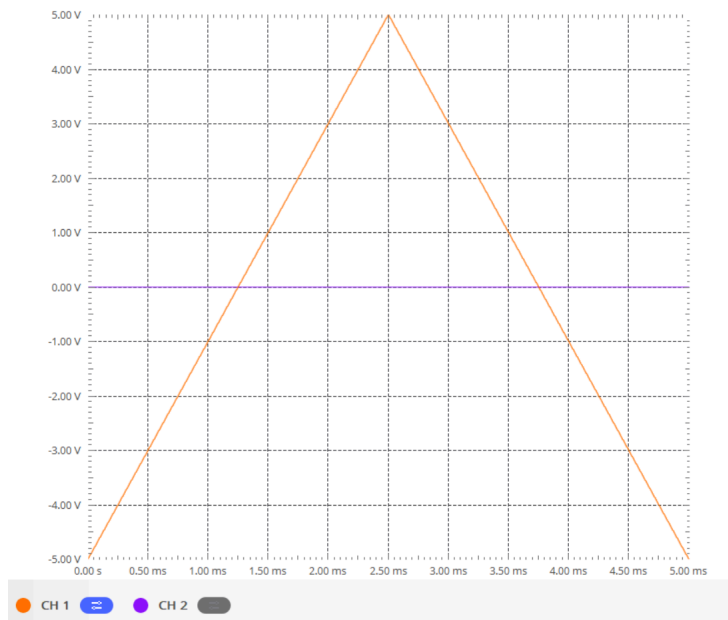
Measurement:



$R1 = 470\Omega$

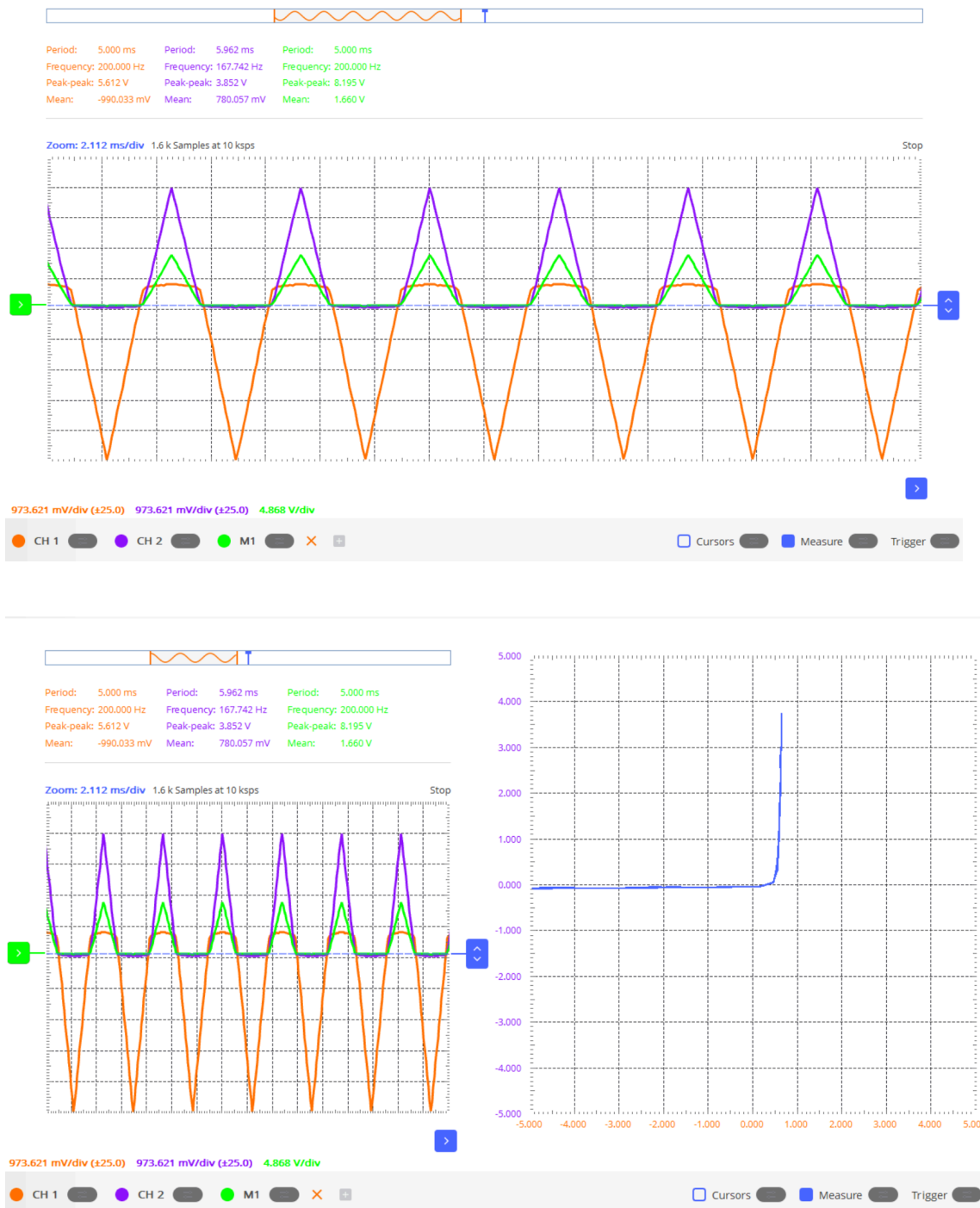
D1 is the 1N914 Diode

Having these settings for the signal generator:



The oscilloscope results in this:

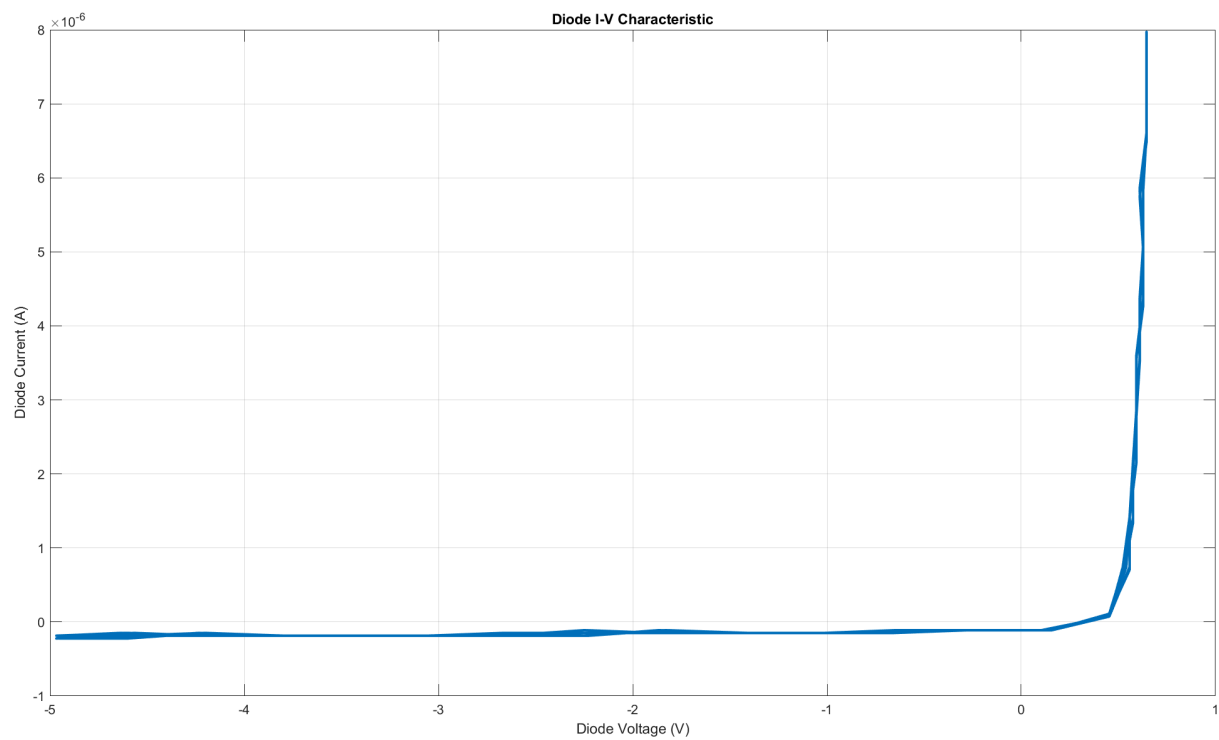
M1 plots the current through the resistor (Math channel with $(VR1/470)*1000$ for visibility)



Program on MATLAB:

```
lab_2_pos_2_graph.m  +
1      M1V = M1V./1000
2      plot(CH1V,M1V,LineWidth=2)
3      hold on
4      xlabel('Diode Voltage (V)')
5      ylabel('Diode Current (A)')
6      grid on
7      title('Diode I-V Characteristic')
```

Resulting graph on MATLAB:



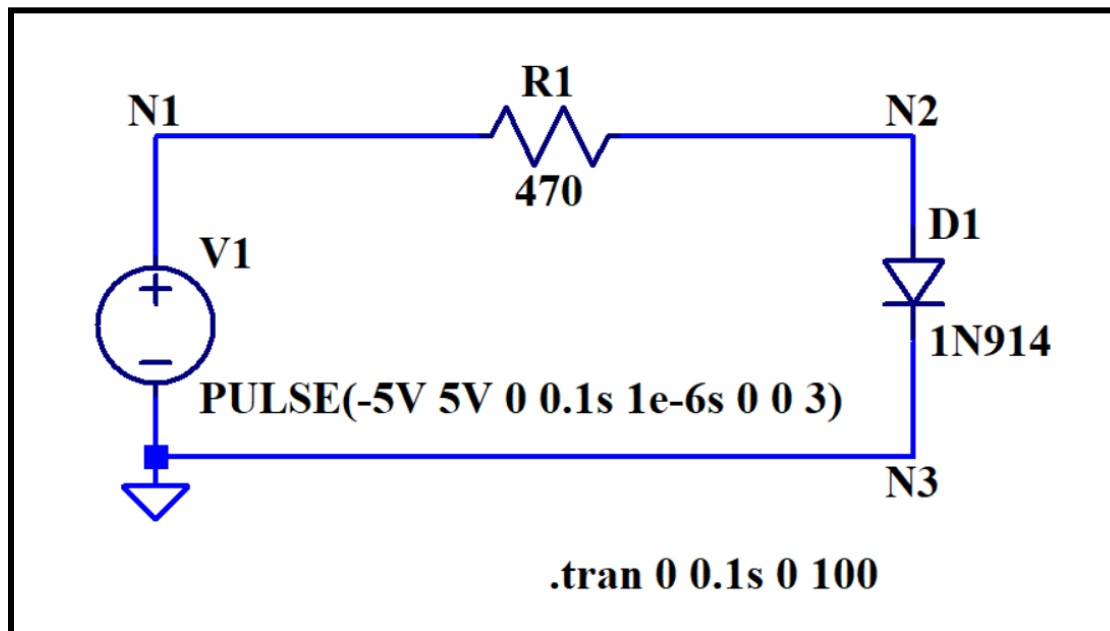
Discussion (and answer related questions in Alpha Lab):

	Analysis	Simulation	Measurement
Shape of Graph	An exponentially increasing graph that rises when Voltage is around 0.5 V	An exponentially increasing graph that rises when Voltage is around 0.5 V	An exponentially increasing graph that rises when Voltage is around 0.6 V.

As shown above, the results of the IV graphs from all measurement methods are accurate and close to each other. According to the measurements, the forward voltage is located at around 0.5 V. The first differential resistance is very close to zero as the slope on the IV curve at the point is very high, close to infinity. The second differential resistance is about 300 V/A as the slope at that point of the IV curve is about 3.3 mA/V. These values are all consistent across the measurement methods, so they would be viable to use in scientific research as the error rate is not high. A possible source of some errors is that real diodes in our parts kit are only sometimes perfectly ideal and can introduce errors into the real circuits.

3. (A) Show/demonstrate that the differential resistance changes in different regions in the diode I-V curve.

Building Block:



Description:

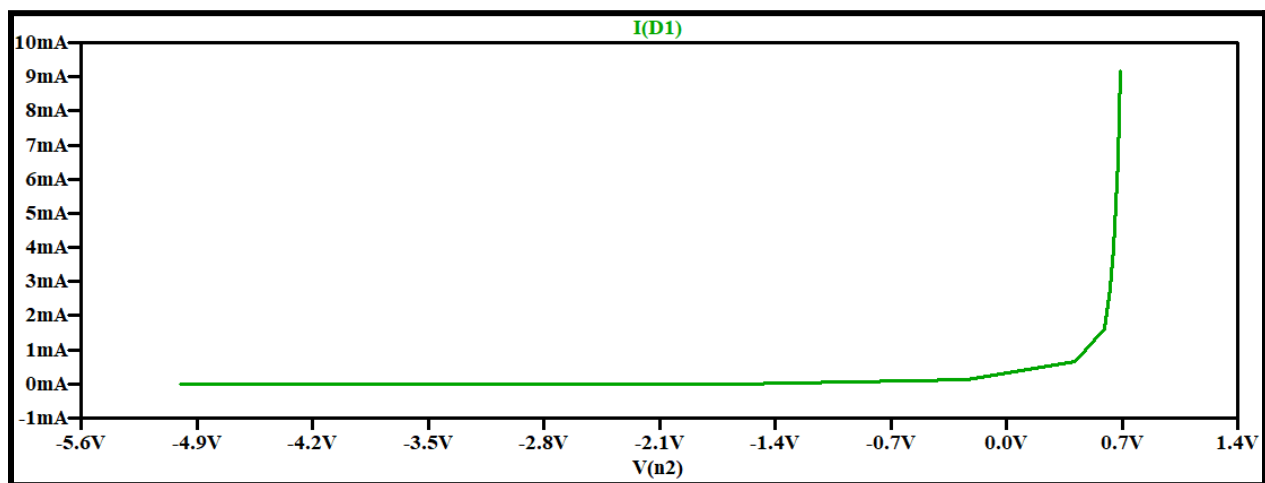
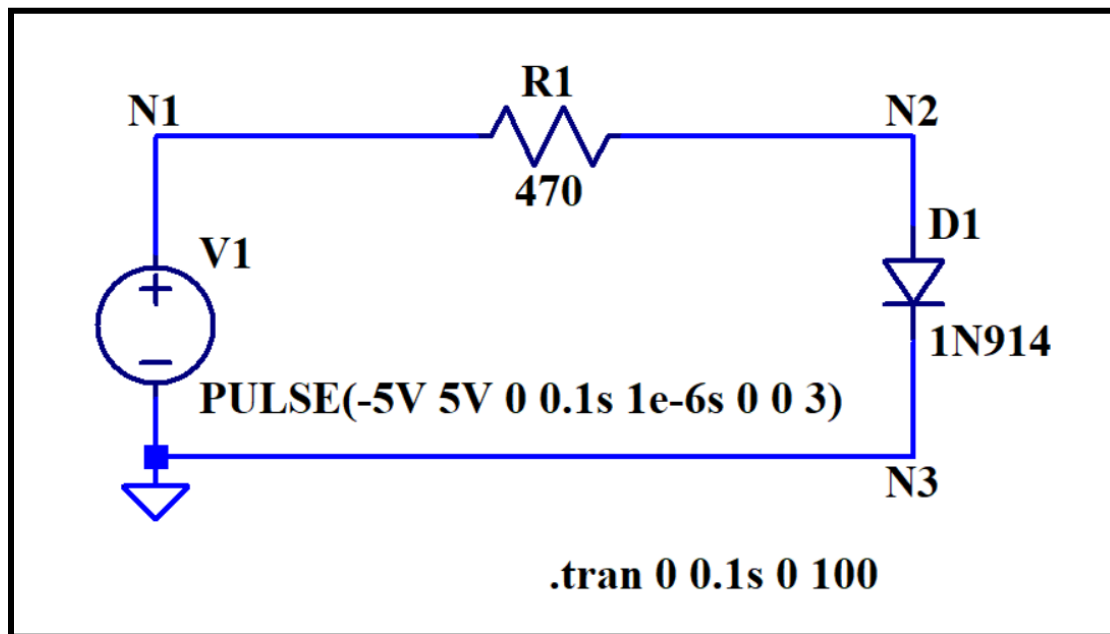
In order to prove that the differential resistance changes in different regions of the diode IV graph, we will look at the simulated and experimental graphs for them. The IV graphs have 3 distinct regions, the reverse-breakdown, the reverse-bias, and the forward. The differential resistance is the inverse slope of the line on the IV graph at certain points, in other words the inverse of the derivative. We will show that on the IV graphs, the differential resistances in the reverse-breakdown region is initially very low and then increases until it reaches the reverse-bias region where it is constantly very high, and then in the forward region where it initially starts high and then decreases until it is very low.

Analysis:

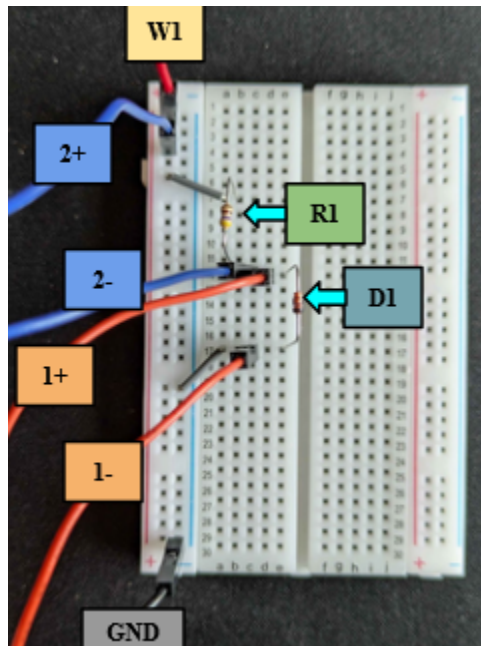
Equation and short description.

Clearly describe how you are applying the concept to your circuit. You should calculate specific values to compare to simulations and experiment.

Simulation:



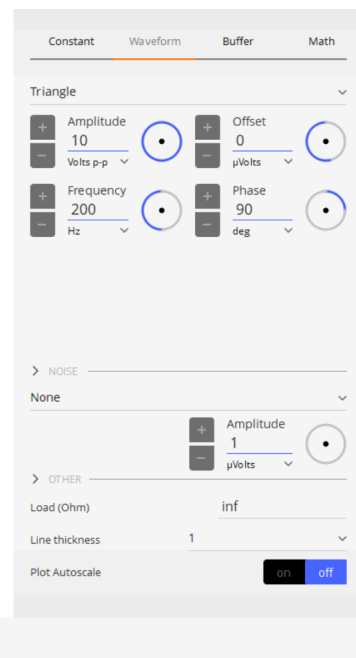
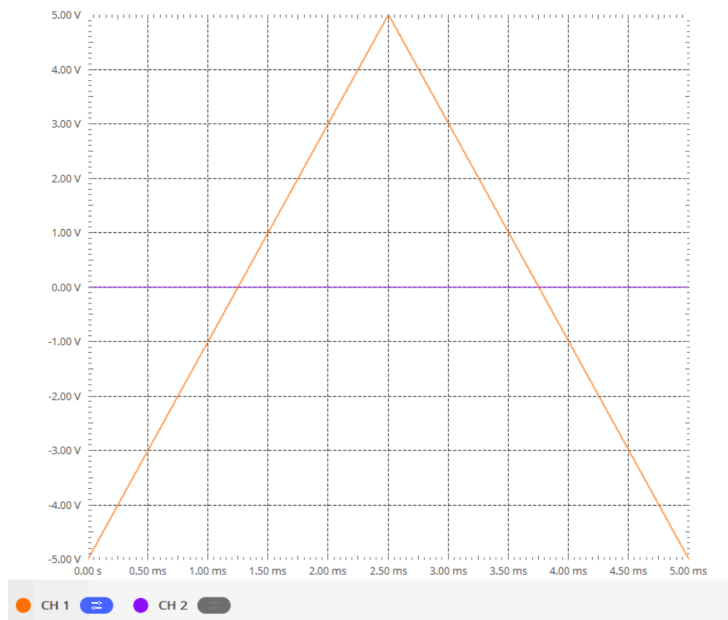
Measurement:



$R1 = 470\Omega$

D1 is the 1N914 Diode

Having these settings for the signal generator:



The oscilloscope results in this:

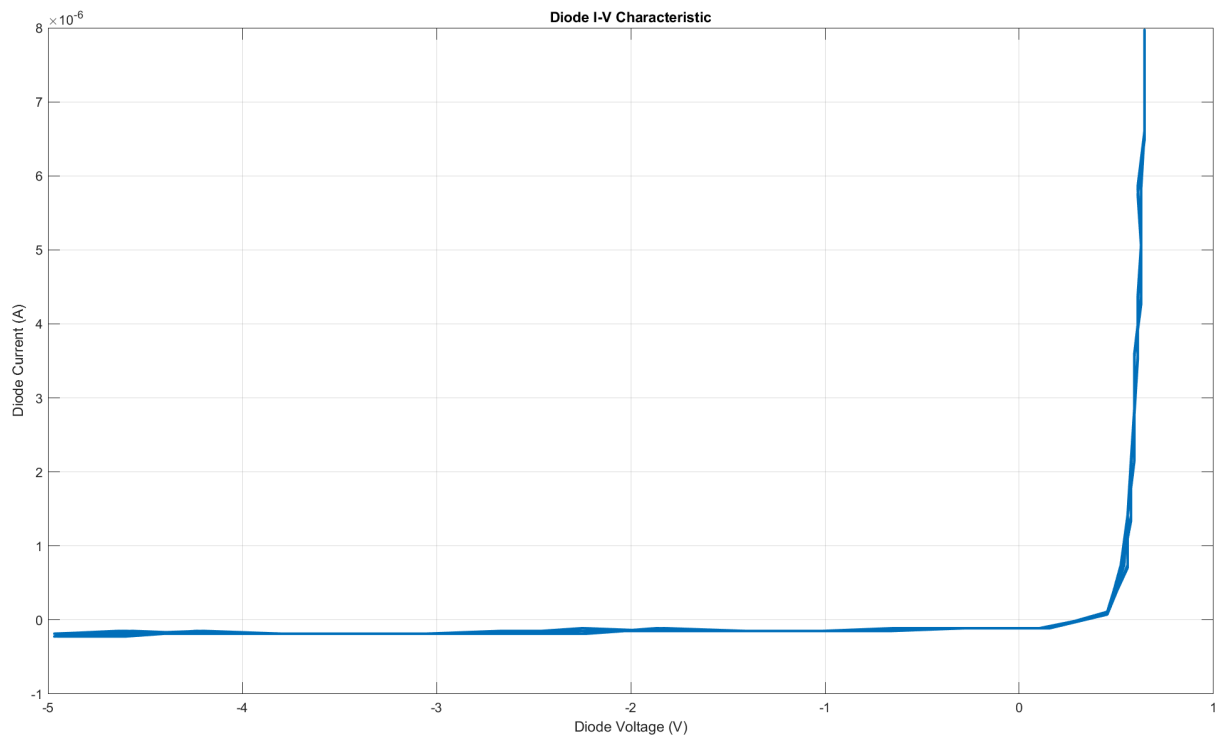
M1 plots the current through the resistor (Math channel with $(VR1/470)*1000$ for visibility)



Program on MATLAB:

```
lab_2_pos_2_graph.m x +
1      M1V = M1V./1000
2      plot(CH1V,M1V,LineWidth=2)
3      hold on
4      xlabel('Diode Voltage (V)')
5      ylabel('Diode Current (A)')
6      grid on
7      title('Diode I-V Characteristic')
```

Resulting graph on MATLAB:



Discussion (and answer related questions in Alpha Lab):

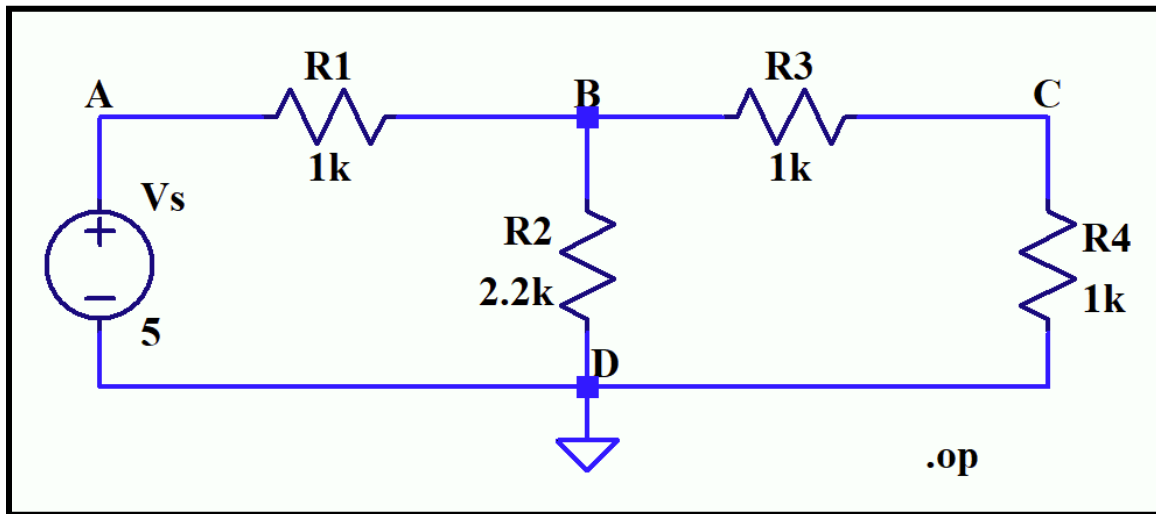
New discussion here.

	Differential Resistance in Simulation Graph	Differential Resistance in Measurement Graph	Differential Resistance in MATLAB Graph
Reverse Bias Region	$1/(\text{Near } 0) = \text{Very High Resistance}$	$1/(\text{Near } 0) = \text{Very High Resistance}$	$1/(\text{Near } 0) = \text{Very High Resistance}$
Beginning Forward Bias Region	$1/0.0075 = 133.33$	$1/0.007 = 142.9$	$1/0.003 = 375$
End Forward Bias Region	$1/(\text{Very High}) = \text{Very Low Resistance}$	$1/(\text{Very High}) = \text{Very Low Resistance}$	$1/(\text{Very High}) = \text{Very Low Resistance}$

Displayed are the differential resistances of the IV Curve at different regions of the curve for the different methods of measurement. All 3 methods show a very high differential resistance in the reverse bias region, very low differential resistance at the end of the forward bias region, and a differential resistance in the low 100s in the beginning of the forward bias region. Although the differential resistance at the beginning of the forward bias region was measured by MATLAB, the result is not different enough to dismiss as it is still in the low hundreds and in the grand scheme of all resistances, isn't too far away. This error may have been caused by a number of things such as MATLAB not having a smooth curve to display the results or a slight error in the estimate of the slope at the point. The concept of the differential resistance changing at different points of the IV curve of a diode is very clear to see and is therefore proved by our measurements.

4. (B) Prove that nodal analysis solves unknown nodal voltages in a circuit

Building Block:



Description:

Use the KCL formula to solve for unknown node values in order to solve for their voltage through the use of matrices. Then compare the calculated values to simulated and measured values to prove how nodal analysis can accurately solve for the unknown voltages.

Analysis:

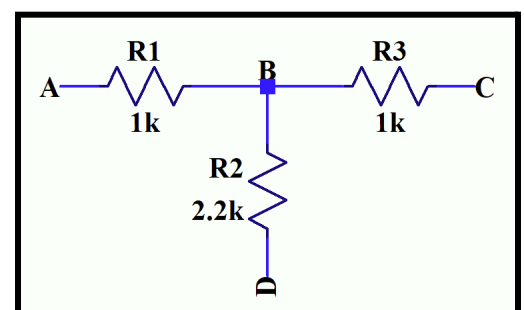
To find # of unknowns:

$$4 \text{ nodes} - 1(\text{source, node A}) - 1(\text{ground, node D}) = 2 \text{ unknowns (node B, node C)}$$

Use KCL in order to create formulas for each node:

Node B:

$$I_1 + I_2 + I_3 = \frac{V_B - V_A}{R_1} + \frac{V_B - V_D}{R_2} + \frac{V_B - V_C}{R_3} = 0$$

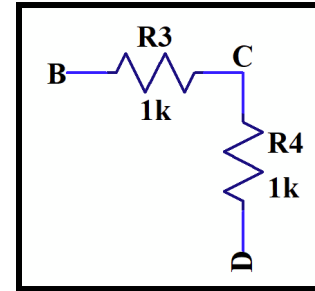


$$\Rightarrow \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)V_B + \left(-\frac{1}{R_3}\right)V_C = \frac{V_A}{R_1} = \frac{5V}{1000\Omega} = 0.005A$$

Node C:

$$I_3 + I_4 = \frac{V_C - V_B}{R_3} + \frac{V_C - V_D}{R_4} = 0$$

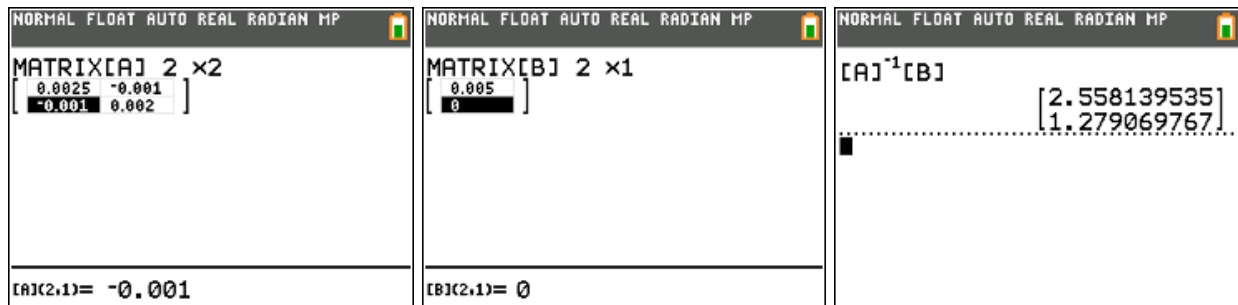
$$\Rightarrow \left(-\frac{1}{R_3}\right)V_B + \left(\frac{1}{R_3} + \frac{1}{R_4}\right)V_C = \frac{V_D}{R_4} = \frac{0V}{1000\Omega} = 0A$$



Put into a matrix to solve for V_B and V_C :

$$\begin{bmatrix} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) & \left(-\frac{1}{R_3}\right) \\ \left(-\frac{1}{R_3}\right) & \left(\frac{1}{R_3} + \frac{1}{R_4}\right) \end{bmatrix} \begin{bmatrix} V_B \\ V_C \end{bmatrix} = \begin{bmatrix} 0.005 \\ 0 \end{bmatrix}$$

To solve the matrix, use calculator (TI-84):

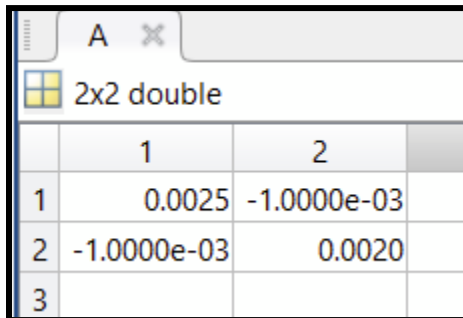


Resulting in $V_B = 2.558V$ and $V_C = 1.279V$

To solve the matrix, on MATLAB:

Create matrix A:

```
>> A = [ ((1/1000)+(1/2200)+(1/1000))  -(1/1000);  
        -(1/1000)  ((1/1000)+(1/1000)) ]
```

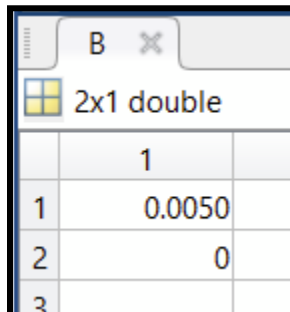


A screenshot of the MATLAB variable viewer showing matrix A. The window title is 'A'. It is a 2x2 double matrix. The values are: Row 1: 0.0025, -1.0000e-03; Row 2: -1.0000e-03, 0.0020.

	1	2
1	0.0025	-1.0000e-03
2	-1.0000e-03	0.0020

Create matrix B:

```
>> B = [ (5/1000); 0 ]
```



A screenshot of the MATLAB variable viewer showing matrix B. The window title is 'B'. It is a 2x1 double matrix. The values are: Row 1: 0.0050; Row 2: 0.

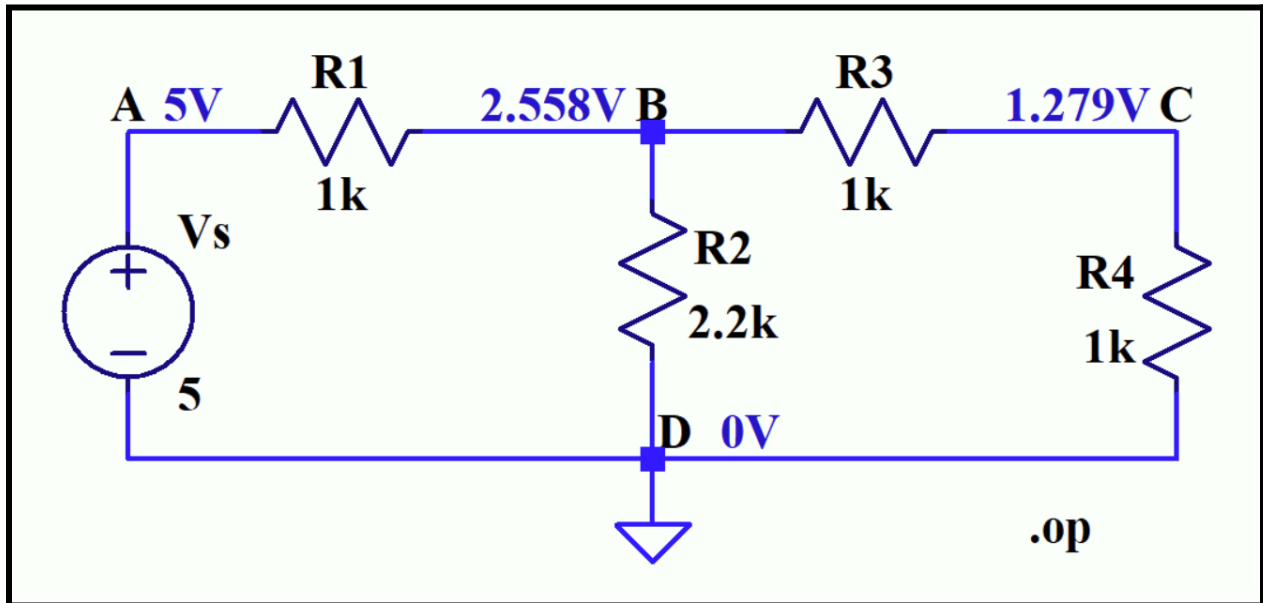
	1
1	0.0050
2	0

And then calculating $A^{-1}B$:

```
>> inv(A)*B  
  
ans =  
  
    2.5581  
    1.2791
```

MATLAB calculates that $V_B = 2.5581V$ and $V_C = 1.2791V$

Simulation:



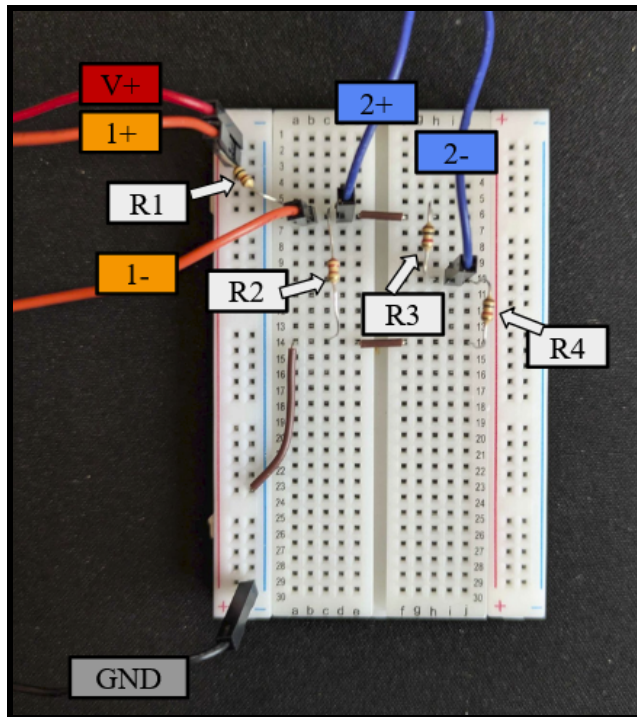
```
* C:\Users\kimbe\Downloads\Draft2.asc

--- Operating Point ---

V(a) :          5          voltage
V(b) :      2.55814      voltage
V(c) :      1.27907      voltage
I(R1) :     -0.00244186   device_current
I(R2) :      0.00116279   device_current
I(R3) :     -0.00127907   device_current
I(R4) :     -0.00127907   device_current
I(Vs) :     -0.00244186   device_current
```

Simulating on LT Spice results in voltage at Node A to be 5V, voltage at Node B to be 2.558V, voltage at Node C to be 1.279V and voltage at Node D (Ground) to be 0V

Measurement:



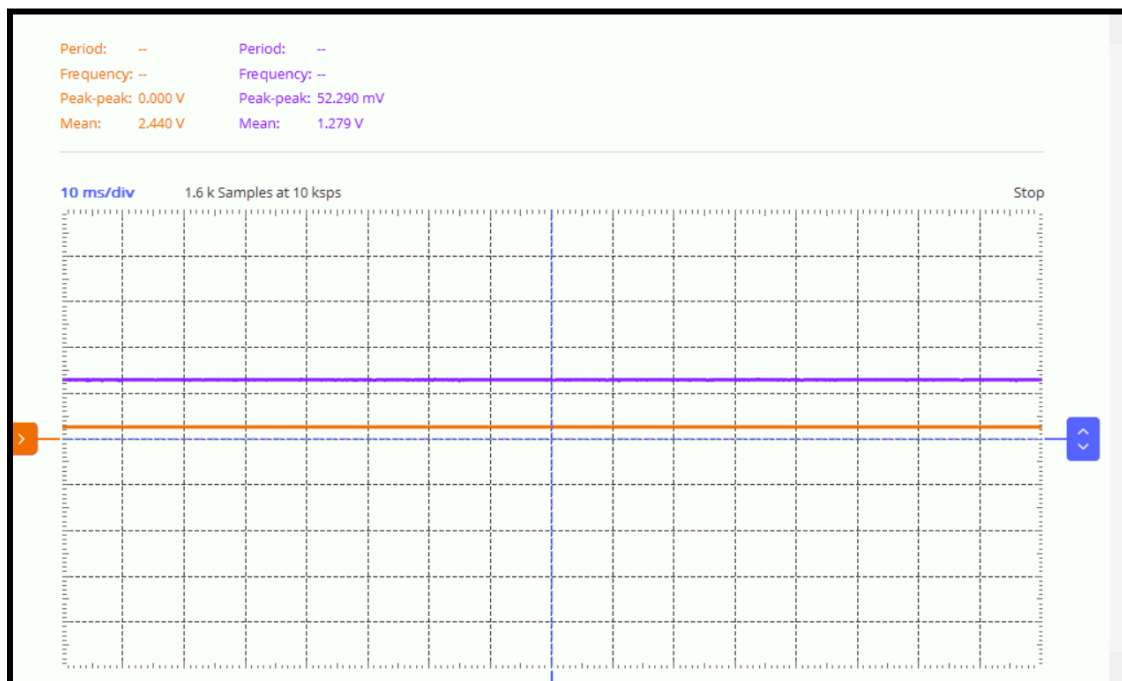
$$R_1 = 1k\Omega$$

$$R_2 = 2.2k\Omega$$

$$R_3 = 1k\Omega$$

$$R_4 = 1k\Omega$$

Using the oscilloscope on Scopy,
CH1 (1+,1-) measures V_B
and CH2 (2+,2-) measures V_C



After turning on the power supply to 5V, the resulting measurements are $V_B = 2.440V$ on CH1 and $V_C = 1.279V$ on CH2

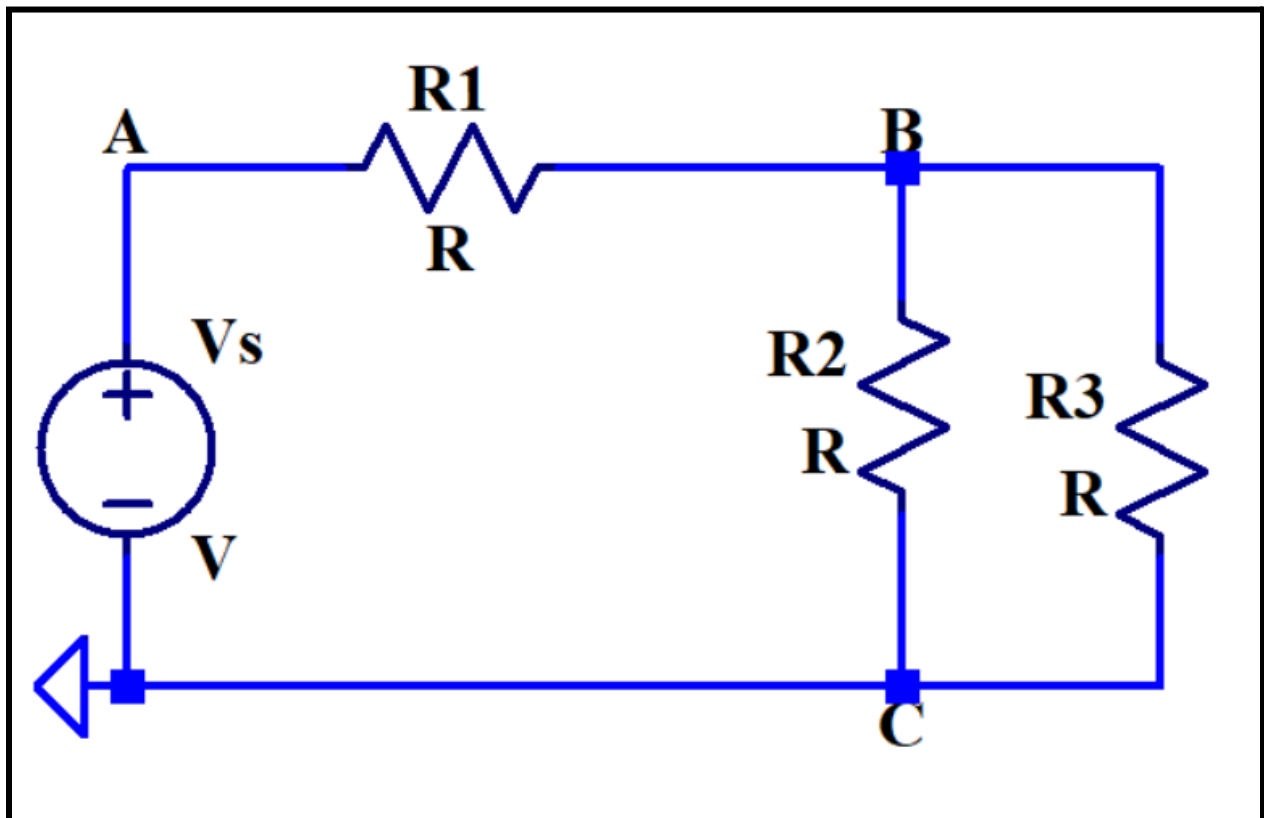
Discussion (and answer related questions in Alpha Lab):

	Analysis	Simulation	Measurement	% Err
V_B (V)	2.558	2.558	2.440	4.61%
V_C (V)	1.279	1.279	1.279	0.00%

These results are pretty much matching the expected results from mathematical analysis and simulation on LT Spice. The voltage at Node B was expected to be 2.558V and was measured to be 2.440V resulting in 4.61% error which is not 0% error but can be resultant from the resistor value not being exactly as they are designed to be and error in the measurement tools that may lose a bit of voltage. The voltage at Node C was expected to be 1.279V and was measured to be 1.279V which exactly matches the value calculated and simulated resulting in a 0% error.

5. (B) Prove/demonstrate your approach to designing a circuit using nodal analysis

Building Block:



Description:

Use circuit analysis in order to find values for resistors and voltage source in order to satisfy the design challenge of having 3 resistors (none in series, two can be in parallel) and one nodal voltage being 2V (not source). In this case, use the circuit layout above to decide values for resistors and source voltage where Node B is where nodal voltage will equal to 2V.

Analysis:

For design challenge #1, we have 1 voltage source and 3 resistors to have a nodal voltage of 2V.

For this circuit, I decided that node B will be 2V and the voltage source will be 3V.

To find the resistor values, use nodal analysis with KCL formula at node B,

$$\text{Node B: } I_{R1} + I_{R2} + I_{R3} = 0$$

$$I_{R1} = \frac{V_B - V_A}{R_1} = \frac{2-3}{R_1} = -\frac{1}{R_1}$$

$$I_{R2} = \frac{V_B - V_C}{R_2} = \frac{2-0}{R_2} = \frac{2}{R_2}$$

$$I_{R3} = \frac{V_B - V_C}{R_3} = \frac{2-0}{R_3} = \frac{2}{R_3}$$

$$-\frac{1}{R_1} + \frac{2}{R_2} + \frac{2}{R_3} = 0$$

Choose $R_1 = 1k\Omega$ so,

$$-\frac{1}{1k} + \frac{2}{R_2} + \frac{2}{R_3} = 0 \Rightarrow 2\left(\frac{1}{R_2} + \frac{1}{R_3}\right) = \frac{1}{1000} \Rightarrow \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{2000}$$

To make things simpler, say that $R_2 = R_3$ so,

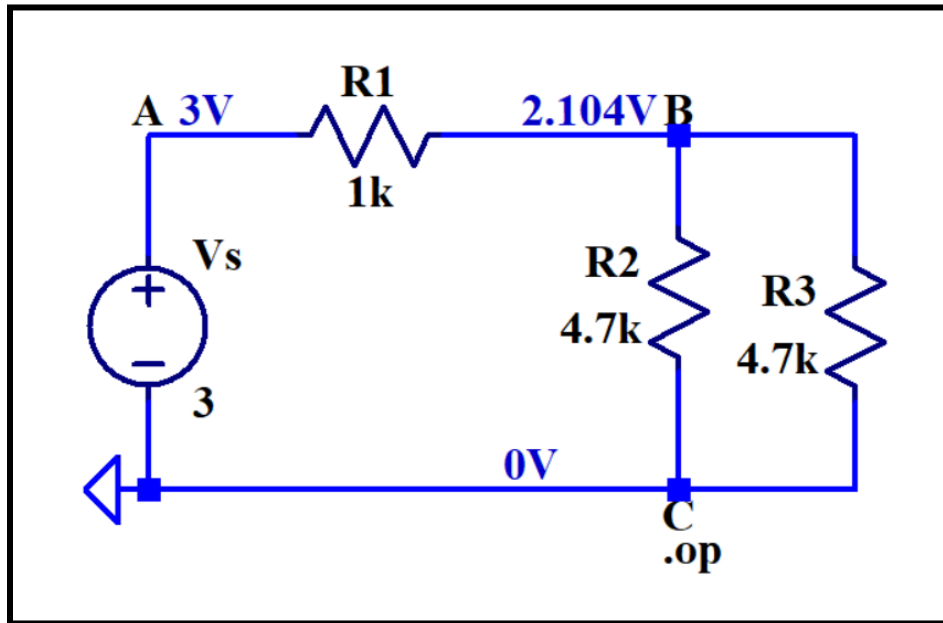
$$\frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{2000} \Rightarrow \frac{2}{R_2} = \frac{1}{2000} \Rightarrow R_2 = 4000 = 4k\Omega$$

Since there is no $4k\Omega$ resistor, use the closest value of $4.7k\Omega$ so $R_2 = R_3 = 4.7k\Omega$

With these resistor values, solve for what V_B will actually come out to be:

$$\frac{V_B - V_A}{R_1} + \frac{V_B - V_C}{R_2} + \frac{V_B - V_C}{R_3} = 0 \Rightarrow V_B = \frac{\left(-\frac{1}{R_1} - \frac{1}{R_2}\right)V_A + \left(-\frac{1}{R_3}\right)V_C}{\left(-\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)} = \frac{\left(-\frac{1}{1000} - \frac{1}{4700}\right)3 + \left(-\frac{1}{4700}\right)0}{\left(-\frac{1}{1000} + \frac{1}{4700} + \frac{1}{4700}\right)} = 2.104V$$

Simulation:



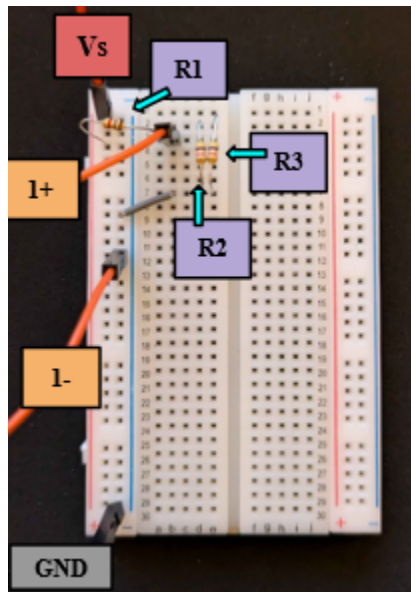
```
* C:\Users\kimbe\Downloads\Draft2.asc

--- Operating Point ---

V(a) :          3          voltage
V(b) :      2.10448      voltage
I(R1) :    -0.000895522  device_current
I(R2) :    -0.000447761  device_current
I(R3) :    -0.000447761  device_current
I(Vs) :    -0.000895522  device_current
```

Simulating the circuit on LT Spice with the values ($V_s = 3V$, $R_1 = 1k\Omega$, $R_2 = 4.7k\Omega$, $R_3 = 4.7k\Omega$) results in the a voltage of 2.104V at Node B.

Measurement:

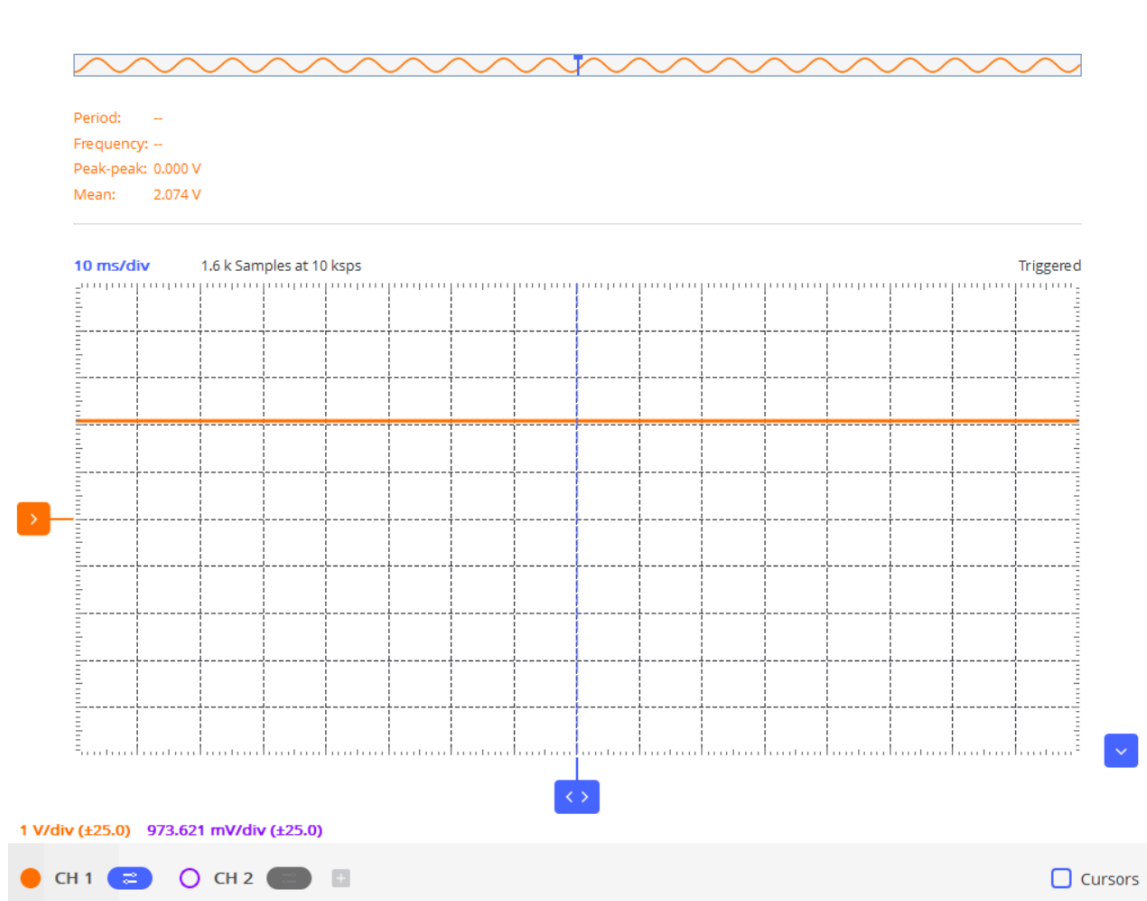


$$R_1 = 1k\Omega$$

$$R_2 = 4.7k\Omega$$

$$R_3 = 4.7k\Omega$$

CH1 measures voltage of Node B, V_B



With a voltage source of 3V, CH1 has a measurement of 2.074V on scopy oscilloscope for V_B

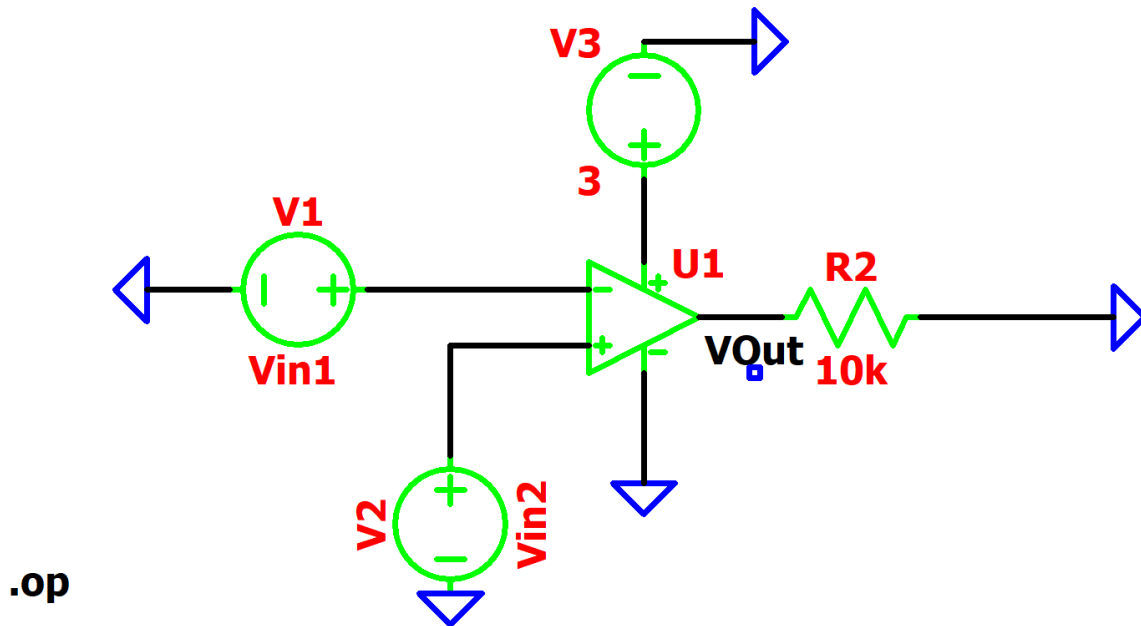
Discussion (and answer related questions in Alpha Lab):

	Analysis	Simulation	Measurement	% Err
V_B (V)	2.104	2.104	2.074	1.43%

The expected value of voltage at Node B was 2.104V according to the design challenge and when simulated on LT Spice with the chosen values, 2.104V was produced as the voltage for Node B. Then through measurement on scope using the oscilloscope, the circuit resulted in a voltage of 2.074V for Node B which is considerably close to the expected value of 2.104V leading to a very small % error of 1.43%. I opted to not find resistances that would lead to exactly 2V since to take a measurement for the voltage, it would be harder to attain the specific resistance values calculated to get exactly 2V. Instead, I chose to use the closest resistance values to the calculated resistance values that were available in the circuit kit. Though it results in V_B not being 2V exactly, it is simpler to model and take measurement from.

6. (C) Prove the function of an op amp comparator

Building Block:



Description:

Short description of the concept you will prove and how you will prove it.

Specify which variables in the circuit will be analyzed.

A comparator is a decision making type of op amp that compares 2 input signals and outputs a supply voltage based on those signals. In the schematic shown above, the 2 signals that will be compared are V1 and V2. Since V2 is connected into the positive terminal of the op amp, if it is

larger than the signal of V_1 , the positive supply voltage of 3 V will be the output voltage, V_{out} . Since V_1 is connected to the negative terminal of the op amp, if it is larger than the signal of V_2 , the negative supply voltage will be output which is 0 V.

Analysis:

1. When $V_{in1} < V_{in2}$, V_{SS+} is V_{out}
2. When $V_{in1} > V_{in2}$, V_{SS-} is V_{out} .

Case 1:

- $V_{in1} = 2.2$
- $V_{in2} = 3.8$

Since $V_{in1} < V_{in2}$, $V_{out} = V_{SS+} = 3 \text{ V}$

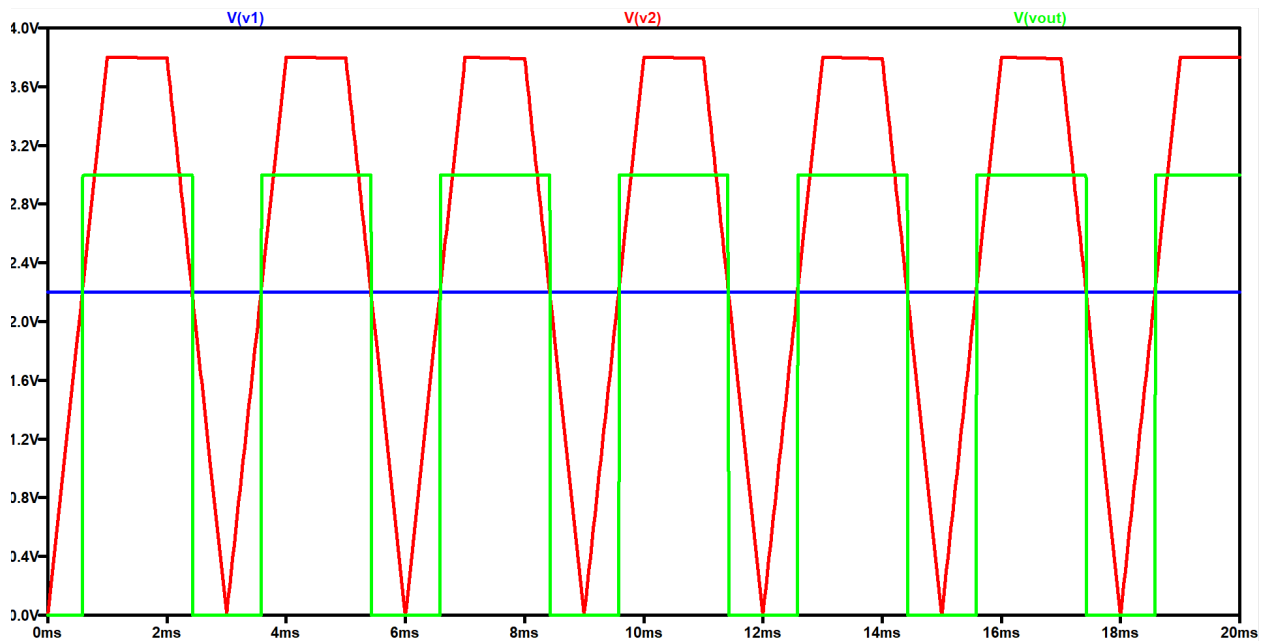
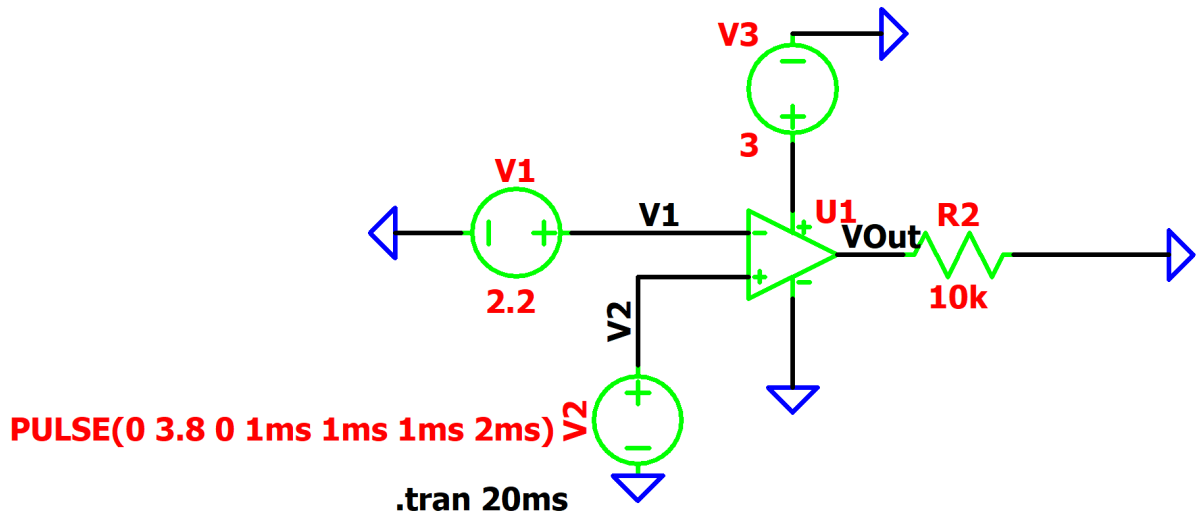
Case 2:

- $V_{in1} = 2.2$
- $V_{in2} = -2.2$

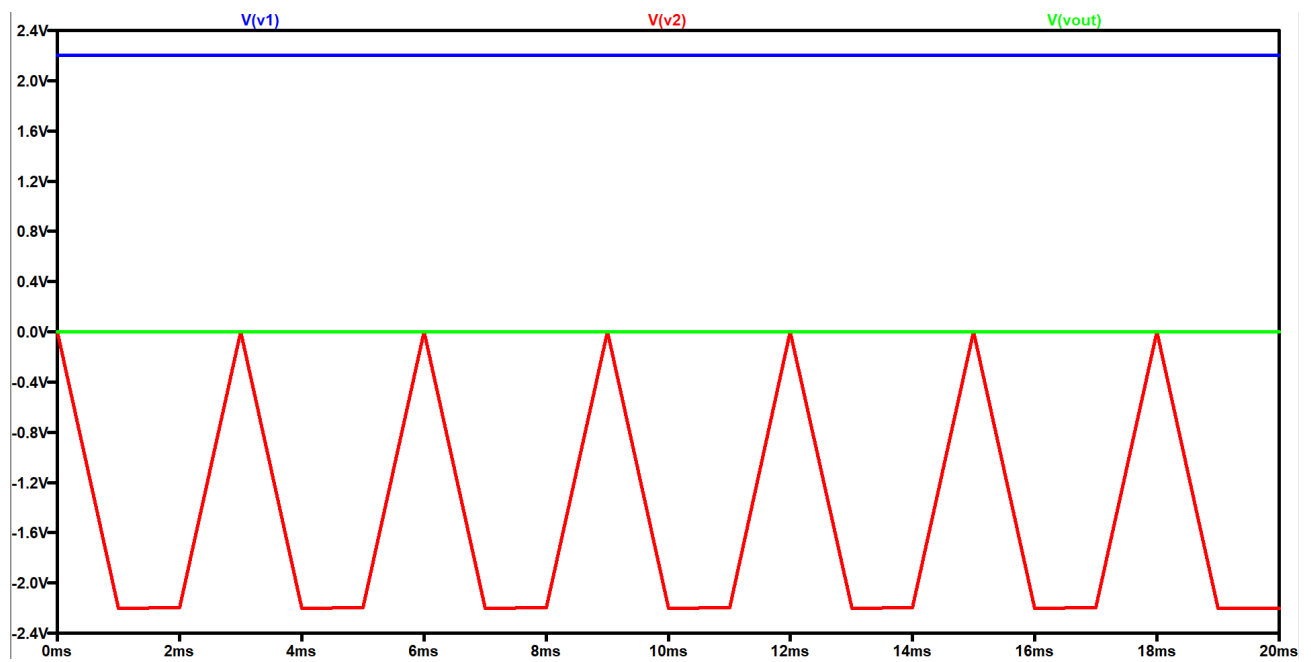
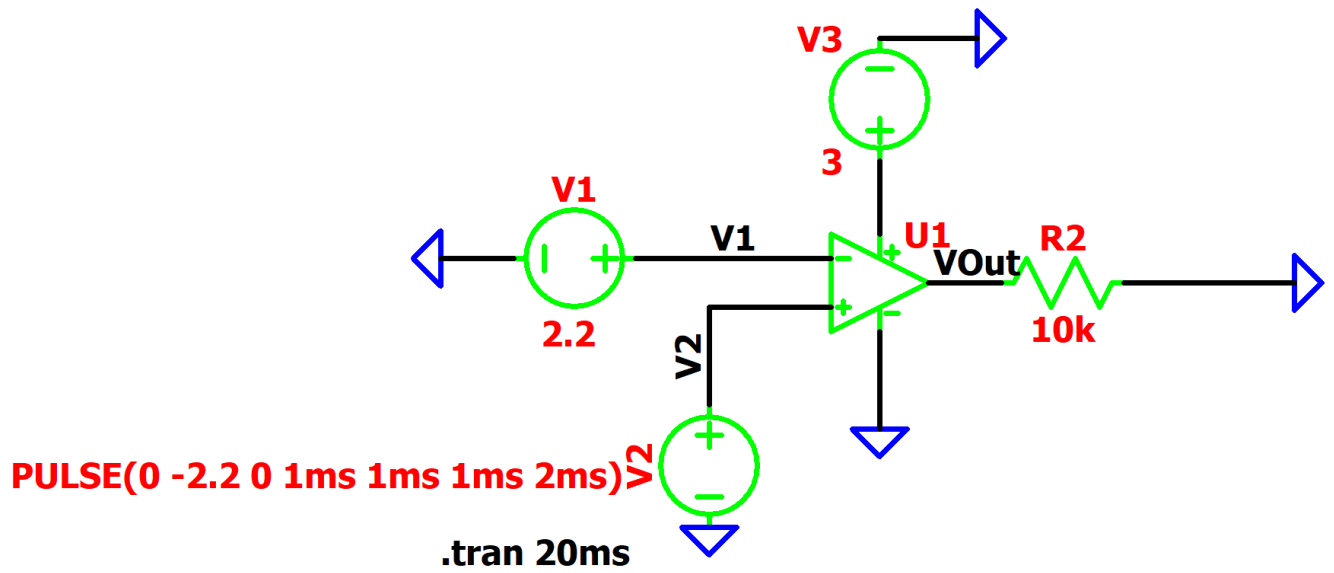
Since $V_{in1} > V_{in2}$, $V_{out} = V_{SS-} = 0 \text{ V}$

Simulation:

$V_{in1} = 2.2\text{ V}$, $V_{in2} = 3.8\text{ V}$



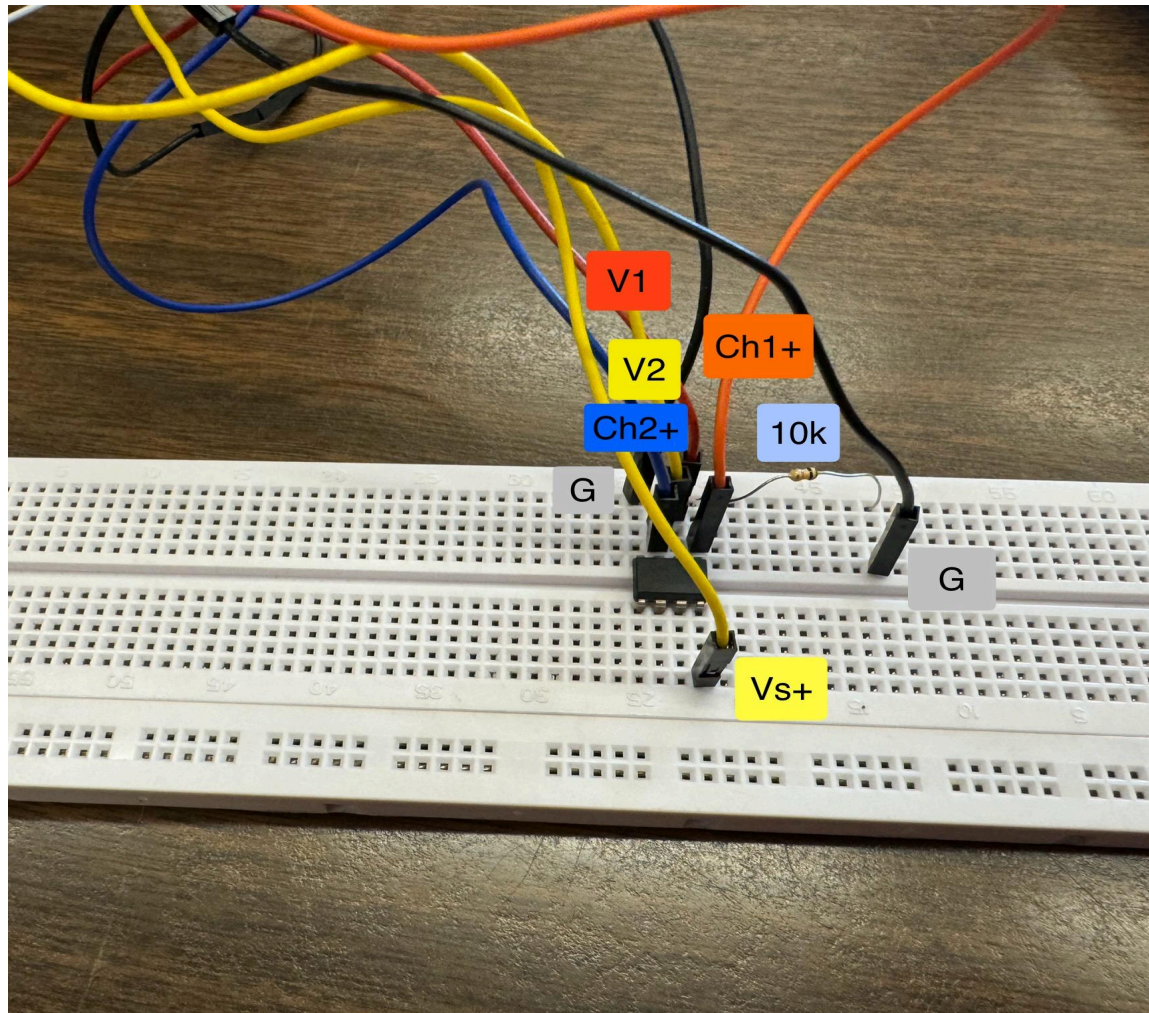
$V_{in1} = 2.2\text{ V}$, $V_{in2} = -2.2\text{ V}$



Measurement:

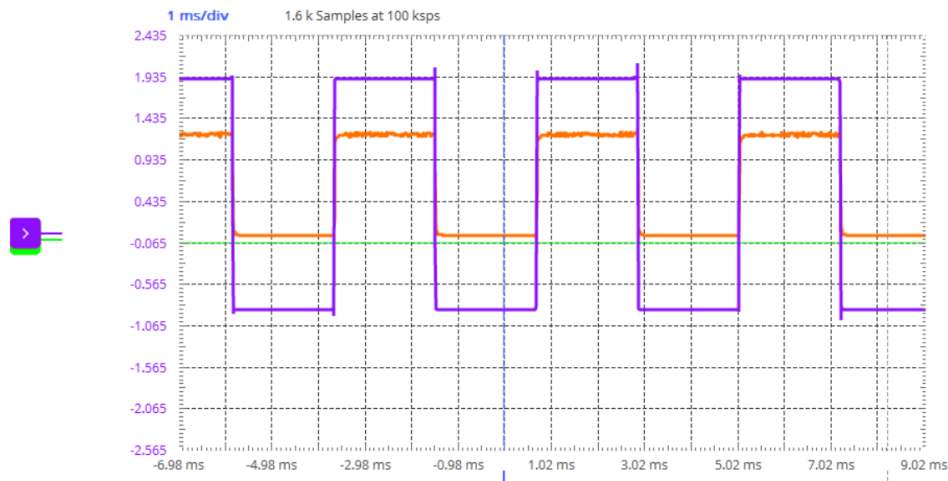
Screenshot of Waveforms output from circuit above and short description.

Remember to clearly show all axes in a measurement plot. Also identify any important portions of the output





CurT1 = 8.213 ms	CurV1 = 2.435 V	Period: --	Period: 4.347 ms	Period: --
CurT2 = 8.213 ms	CurV2 = 2.435 V	Frequency: --	Frequency: 230.061 Hz	Frequency: --
$\Delta t = 0$ s	$\Delta V = 0.000$ V	Peak-peak: 1.386 V	Peak-peak: 3.090 V	Peak-peak: 0.000 V
1/ Δt = Infinity		Mean: 600.800 mV	Mean: 454.882 mV	Mean: 0.000 V



500 mV/div (± 25.0) 500 mV/div (± 25.0) 1 V/div

Channel 2

HORIZONTAL

+

Time Base

1

ms

-

+

Position

1.02

ms

-

VERTICAL

+

Volts/Div

500

mVolts

-

+

Position

65

mVolts

-

SETTINGS

CH Thickness

3

Curve Style

Lines

Memory depth

1600

Probe Attenuation

1

Software

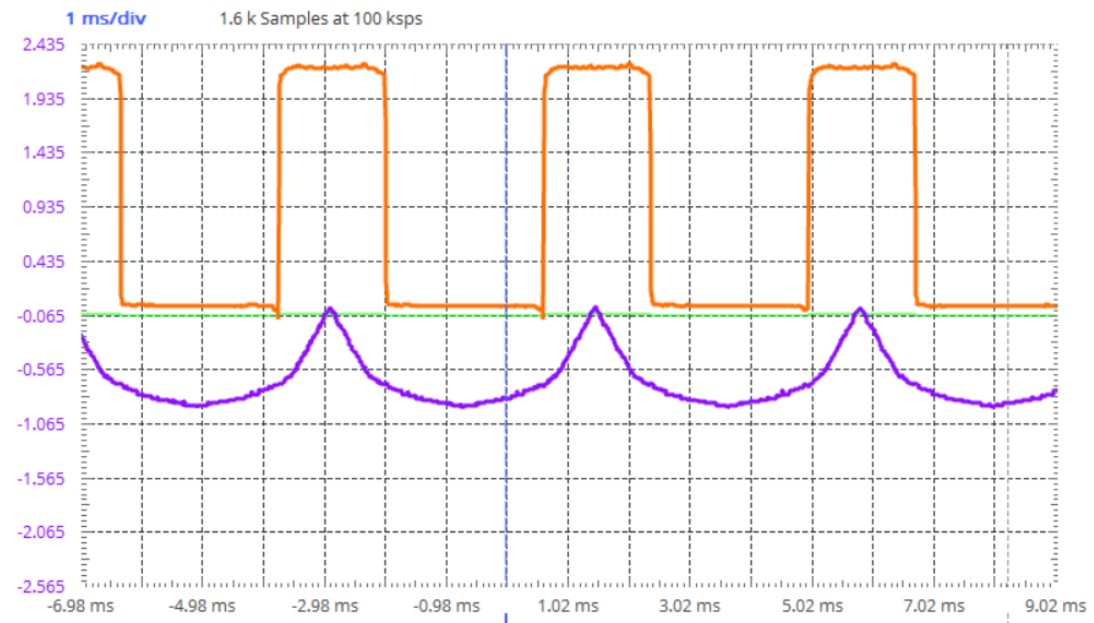
AC Coupling

on

off



CurT1 = 8.213 ms	CurV1 = 2.435 V	Period: --	Period: --	Period: --
CurT2 = 8.213 ms	CurV2 = 2.435 V	Frequency: --	Frequency: --	Frequency: --
$\Delta t = 0$ s	$\Delta V = 0.000$ V	Peak-peak: 2.355 V	Peak-peak: 918.705 mV	Peak-peak: 976.563 μ V
1/ Δt = Infinity		Mean: 827.689 mV	Mean: -683.111 mV	Mean: 360.107 μ V



500 mV/div (± 25.0) 500 mV/div (± 25.0) 1 V/div

Discussion (and answer related questions in Alpha Lab):

Comparison of Analysis, Simulation and Measurement results. Both a simple summary of results (like a numerical chart of values) and a simple description that details if the results are as you expect. Also include any speculation as to why they may be different from one another if they are different. What variation is too much for example...explore this.

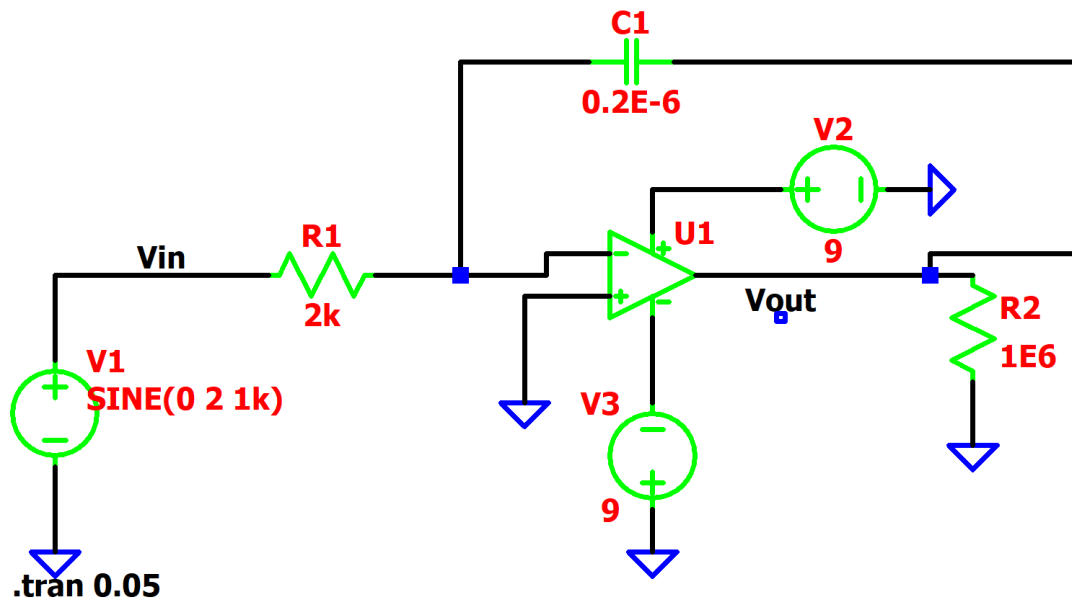
Include a table that compares mathematical, simulation, and experimental results.

	Analysis Vout	Simulation Vout	Measurement Vout
Vin2= 3.8 V	3 V for 2 ms then 0 for 1ms	3 V for 2 ms then 0 for 1ms	1.4 V for 2ms then 0 for 2 ms
Vin2= -2.2 V	0 V the whole time	0 V the whole time	Square wave between 0 V and 2.4 V.

For the comparator schematics, the results are displayed in the table above. Unfortunately, the experimental results did not display how they should have. We thoroughly attempted to change and fix the circuit on the circuit board, but nothing seemed to change the output values. All of the supply voltages and the input voltages were correct, but the output was not. This variation is too much to be taken as useful to compare to the analysis and simulation results. Possible reasons for this large variation is that the amplifier we were using was faulty or that Scopy was misreading the values of the amplifier.

7. (C) Prove the function of a mathematical op amp

Building Block:



Description:

An integrator circuit uses the mathematical operation of integration to integrate a signal's voltage over a given amount of time outputting a signal proportional to the negative integral of the input voltage. To do this, the op amp is the same as an inverting amplifier circuit, but instead of a resistor in the negative feedback loop, a capacitor is used. The capacitor changes the circuit into an RC circuit and provides a time element to the op amp. As time progresses, charge on the capacitor will build up and cause the output voltage to increase linearly if using a DC circuit as the input voltage is inverted to positive.

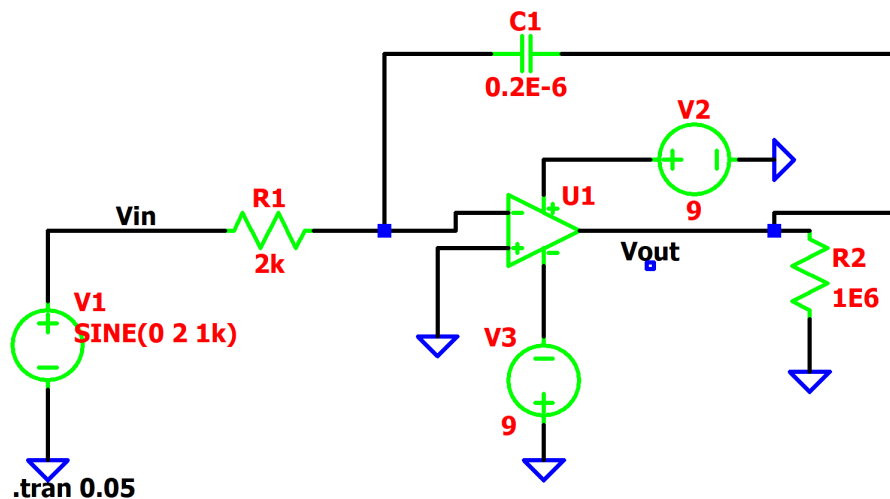
Analysis:

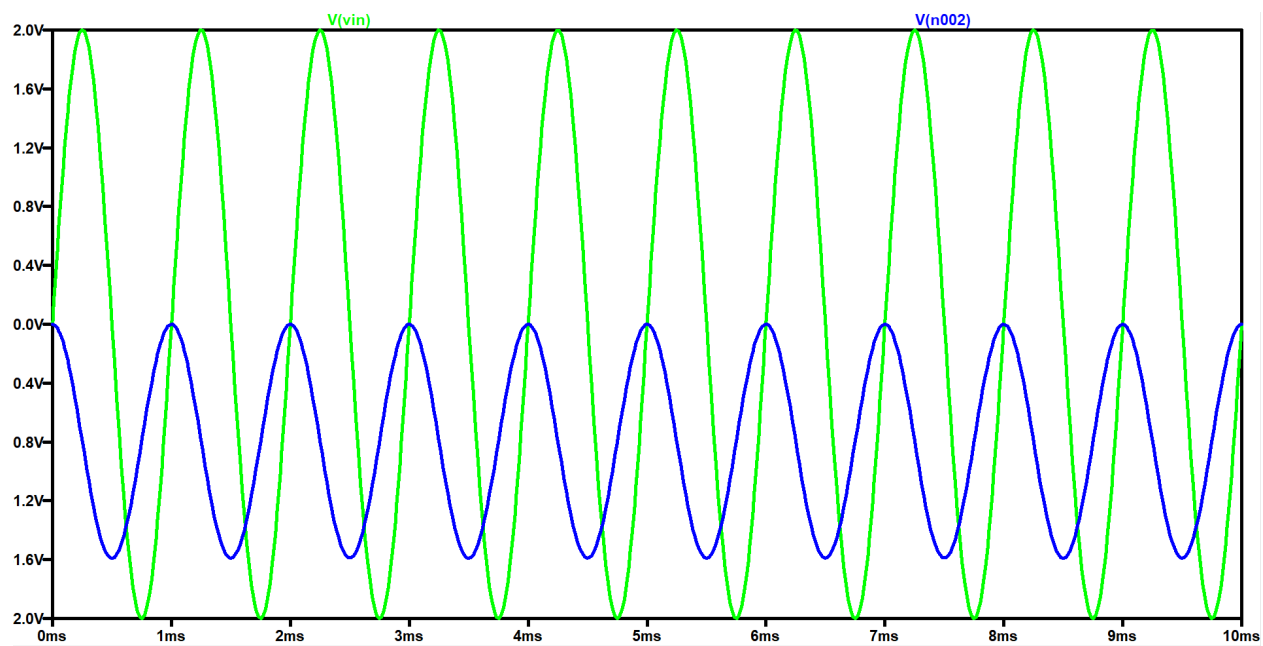
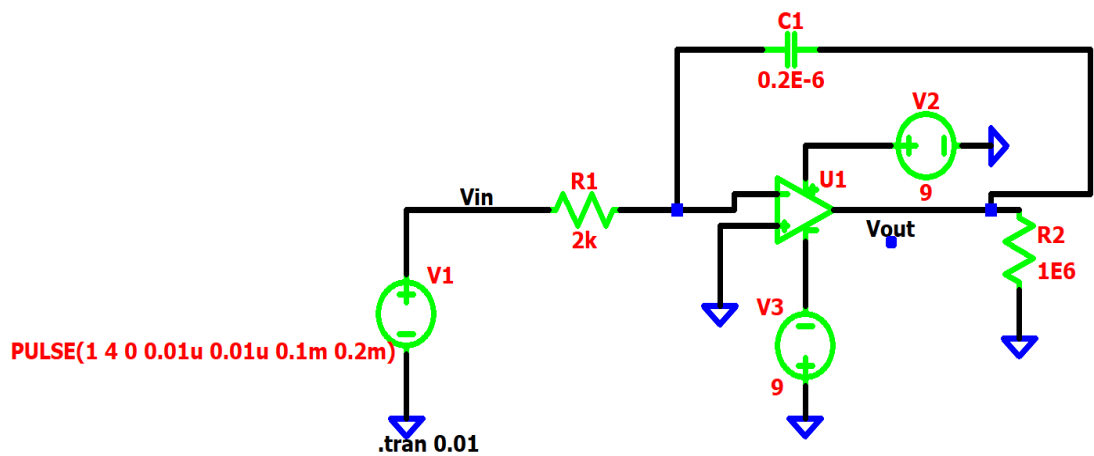
For an integrator circuit, the mathematical calculations are similar to the inverting amplifier circuit, but using a capacitor in the negative feedback loop. At N1, using KCL, the current through R_i will be equal to the current, i_{in} , as the current in the negative port of the op amp is zero. For a capacitor, $i = C(dV/dt)$ where the C is the capacitance of the capacitor at C_f . In this schematic, $i_{in} = d(V_{N1} - V_{out})/dt$ and since $i_{in} = iR_i$, the equation will be $(V_{in} - V_{N1})/R_i = C_f(d(V_{N1} - V_{out})/dt)$. In op amps, the voltage into the negative node will be 0, so this simplifies the equation down to $V_{in}/R_i = C_f(d(-V_{out})/dt)$ as $V_{N1} = 0V$. Now by integrating both sides with respect to time, the equation will turn into $(1/R_i) \int V_{in} dt = -C_f V_{out}$ and then solving for V_{out} will be $V_{out} = (-1/R_i C_f) \int V_{in} dt$.

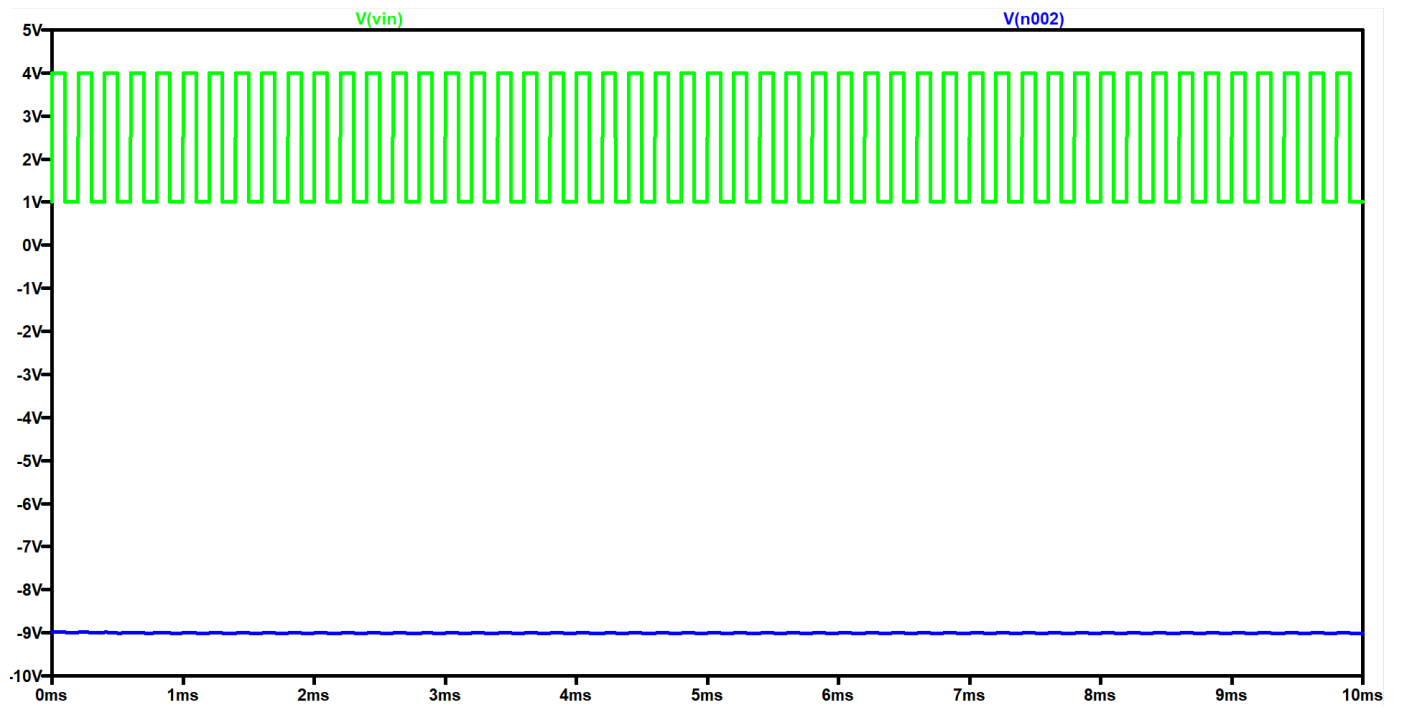
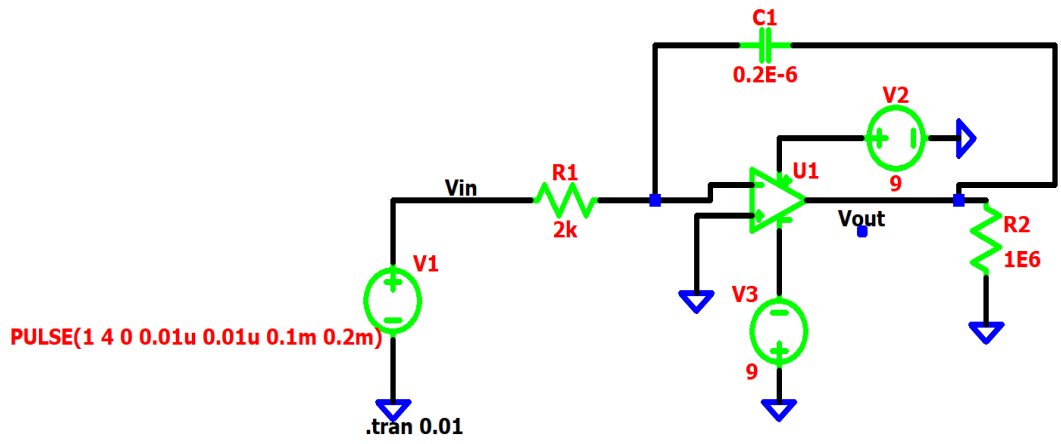
$$V_{out} = (-1/(0.2 \times 10^{-6})(2k)) \int V_{in} dt = -2500 \int V_{in} dt = -2500 \int_0^{0.16ms} V_{in} dt = -9V$$

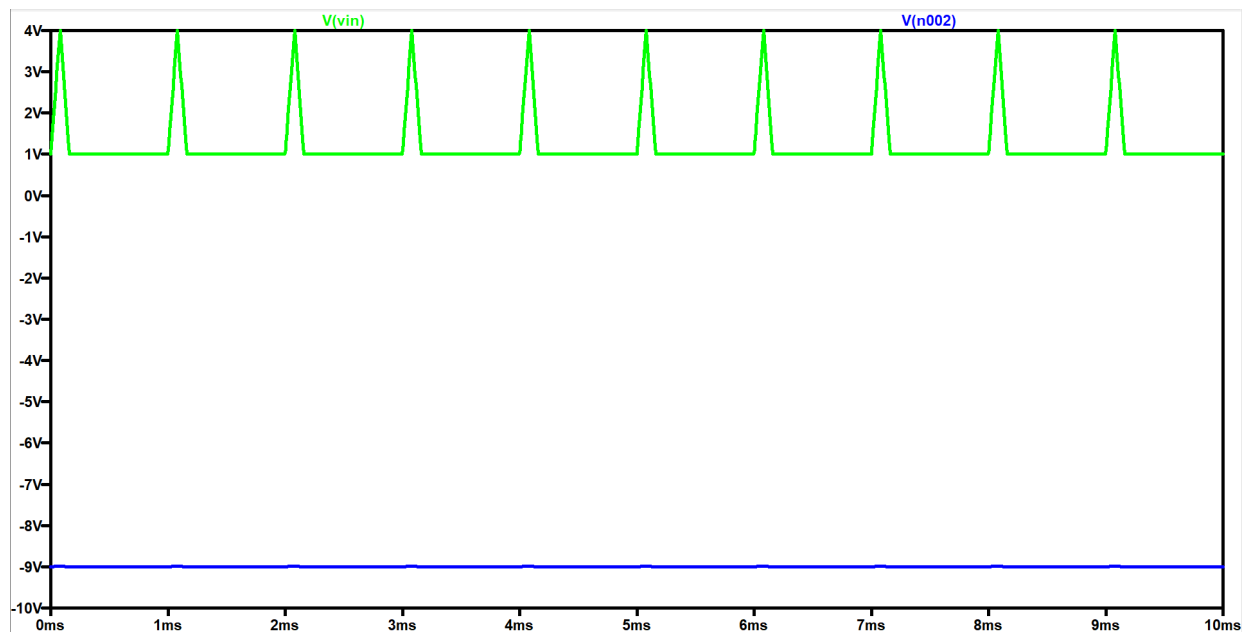
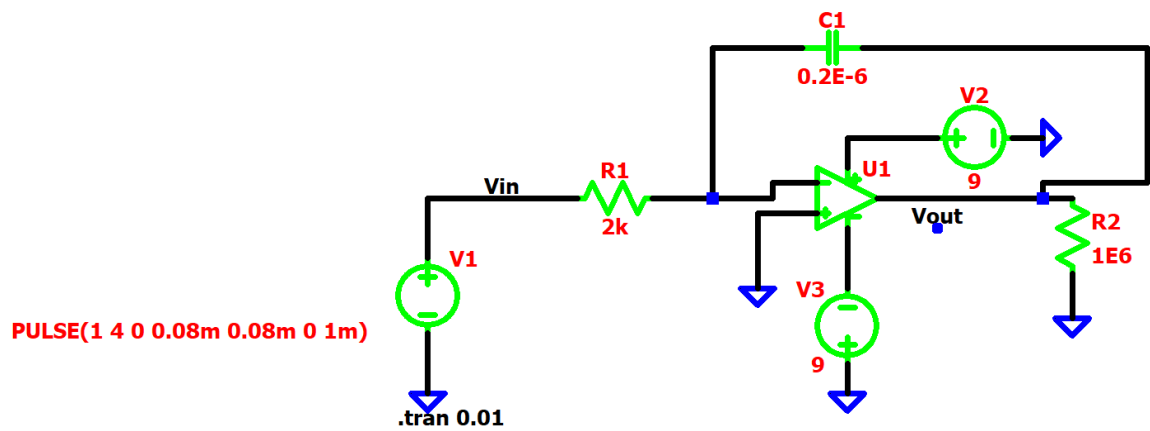
Simulation:

In the schematic shown below, constructed in LTSpice, is an integrator circuit. The voltage source is inputting a sine wave with an amplitude of 2V, a frequency of 1kHz, and with no DC offset. Also, the circuit contains an initial resistor, R_1 , with a resistance of 2k ohms, and an output resistor, R_2 , with a high resistance of 1×10^6 ohms. The capacitor, C_1 , has a capacitance of 0.2×10^{-6} F, and there are $\pm 9V$ voltage supplies. These values are all consistent with the schematic given in the instructions for the integrator circuit. The graph comparing the input voltage's sine wave, V_{in} , and the output voltage's sine wave, V_{out} , are shown in the picture below. Also shown are the schematics with square waves and triangle waves being created. To do this, I just changed the voltage source to a pulsed source and made the jumps go from 1V to 4V, but for the triangle waves I just had the time of ascent and descent both 0.08ms, but in the square waves they were both very small so the drop was almost instantaneous.





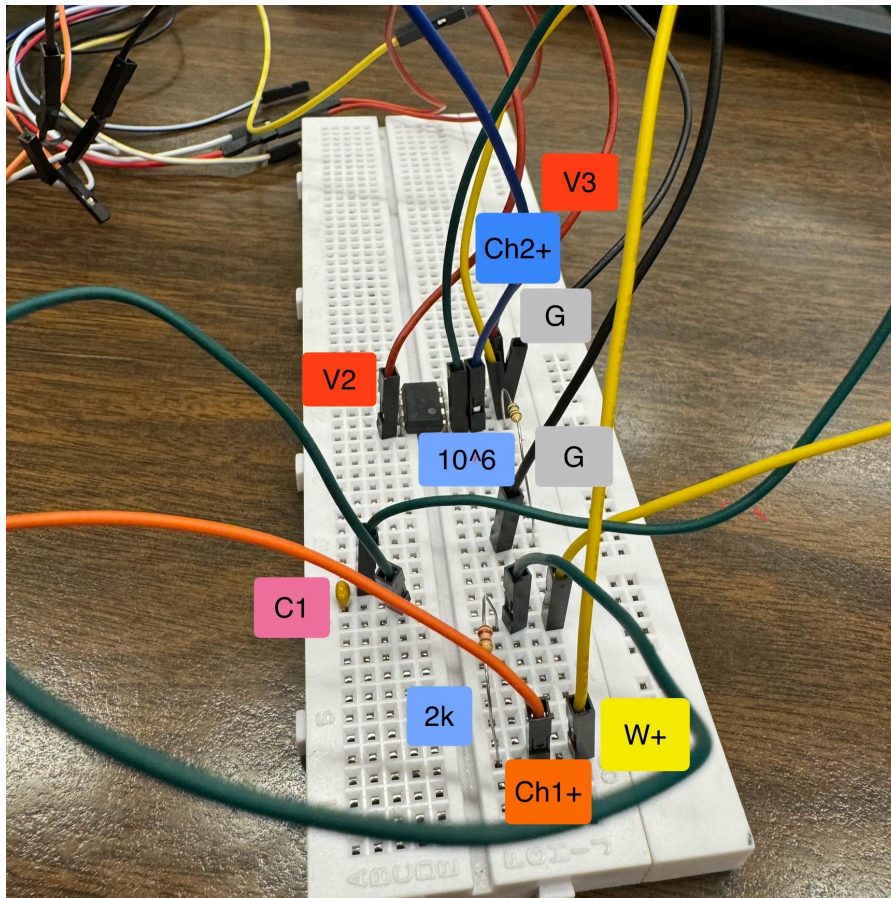


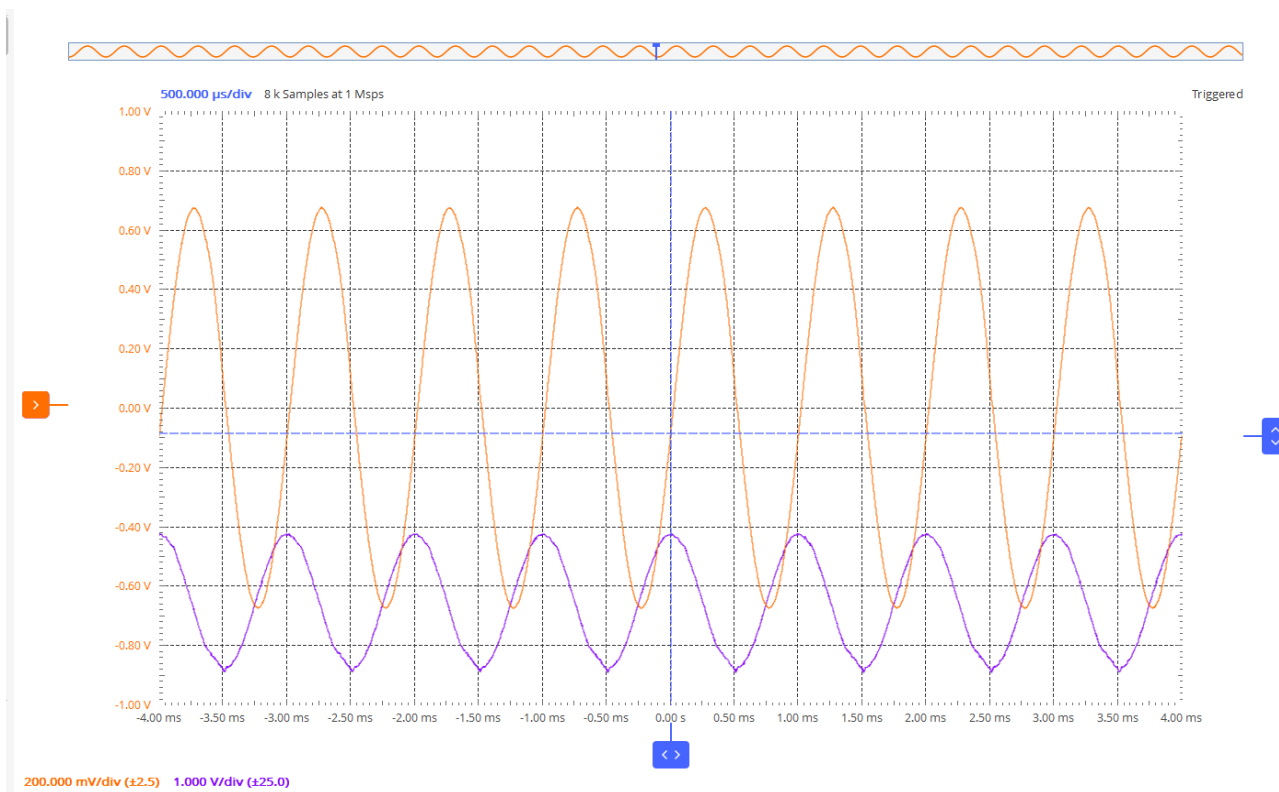
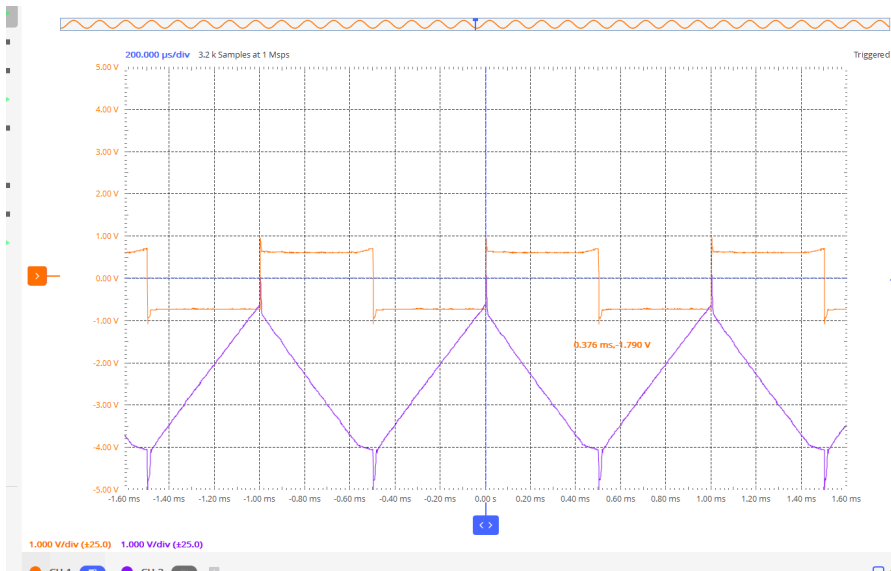


Measurement:

Below is a photo of the circuit that was constructed to integrate various input voltages. The resistor is 100 ohms and the capacitor measures 1uF. The capacitor is fed back from the op97f's output to the inverting input. In the oscilloscopes, the orange line represents the input voltage and the purple line represents the output. The integrator circuit integrates each voltage correctly, producing a triangle wave when a square wave is inputted and producing a sinusoid $\pi/2$ radians out of phase with the input (in other words, a cosine wave is produced when a sine wave is inputted).

$R = 100 \text{ } \Omega$ $C = 1 \mu\text{F}$





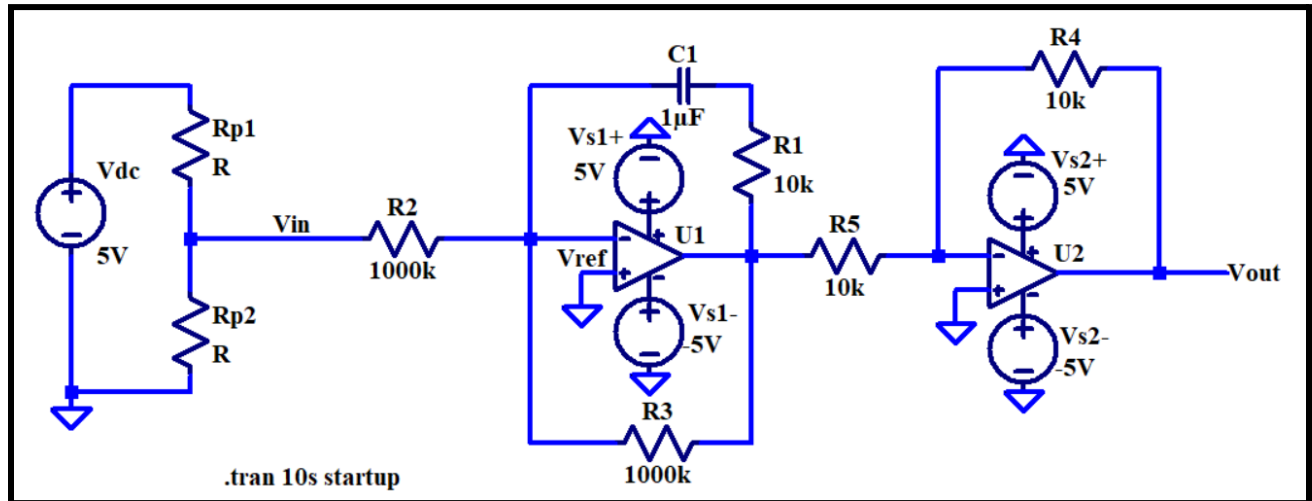
Discussion (and answer related questions in Alpha Lab):

The results table is displayed below. For the measurement, the shape of the output wave was a triangular shaped wave that was not expected. The flat part of the wave where there was no triangular shape, did have a voltage of -9 so this was a correct measurement. For this reason, I don't think the variation is too much where the measurement did not add valuable information about the system. We aren't too sure what caused this odd shape but we think it might have something to do with the way Scopy creates waveforms in its signal generators.

Wave Type	Input Amplitude	Analysis Output	Measurement Output	Simulation Output
Pulse	$4-1=3\text{ V}$	-9 V	-9 V	-9 V

8. (D) PI and PD Controllers

Building Block:



Description:

PI Controller designed with 2 inverting op amps. V_{in} is the input voltage controlled by a 5kΩ potentiometer represented as the voltage divider on the left with a 5V dc source given from an arduino (had to improvise since V_s for op amps occupied $V_{+/-}$ on the instrumentation board) and 2 resistors (R_{p1} and R_{p2}). The voltage sources are +5V and -5V for each op amp. This is a PI controller with the incorporation of the RC in parallel with R_3 . This design results in V_{out} (voltage output) being about equal to V_{in} (input voltage), within around 5% error. The integral controller is C_1 and proportional controller is R_1 .

Analysis:

Voltage divider to model voltage input, non-inverting error op amp

$$V_{in} = V_{dc}(R_{p2}/(R_{p2}+R_{p1}))$$

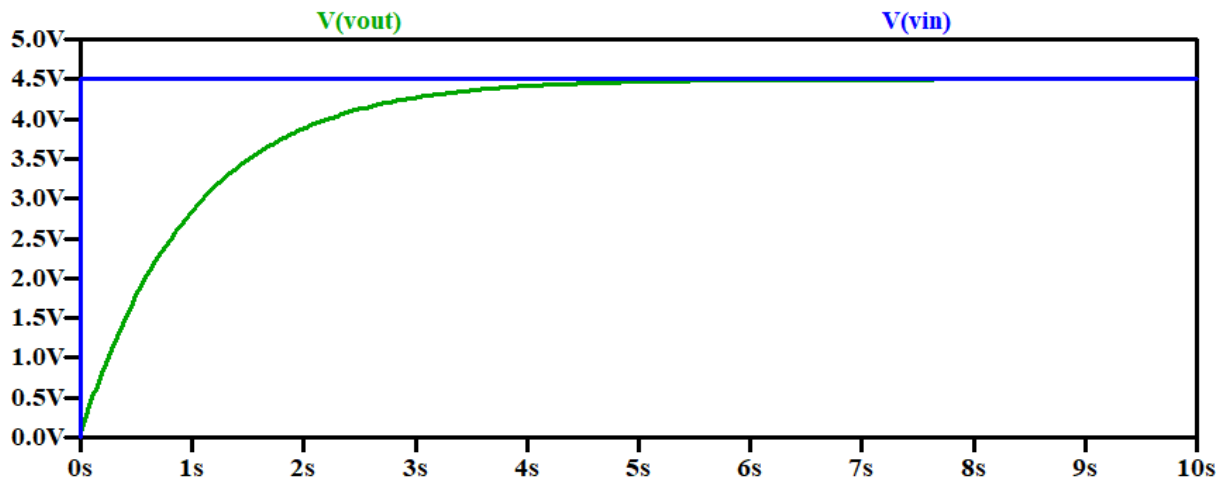
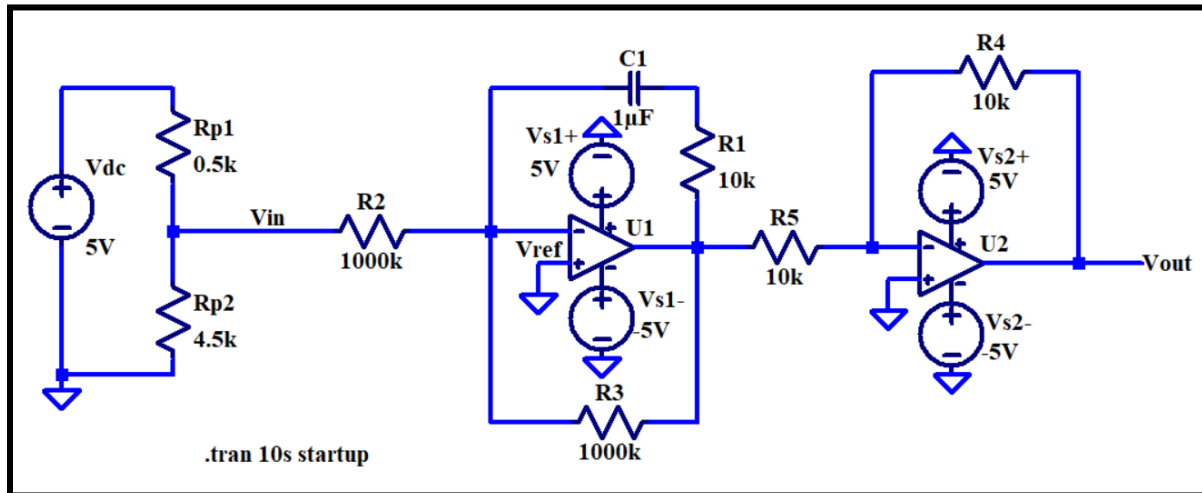
Op Amp 1- Is an integrating op amp with 3 resistors and a capacitor. The capacitor charges up until it reaches steady state where it would appear as an open circuit. This means no current will be flowing through that section of the op amp. This in essence will turn the op amp into an inverting op amp with the resistor R3 as 1000k as time goes on. This means the output voltage of this op amp will be, $V_{out} = -V_{in}(R_3/R_2) = -V_{in}(1000k/1000k) = -V_{in}$. Using this output voltage, the transfer function is calculated as $H = V_{out}/V_{in} = -V_{in}/V_{in} = -1$.

Op Amp 2- This amplifier is an inverting amplifier with two 10k resistances. The output voltage for an inverter is $V_{out} = V_{in}(-R_f/R_i)$. In this inverter, $R_f = R_i$ so the output voltage will just be equal to the negative input voltage. This comes out to a transfer function of -1 as $V_{out}/V_{in} = -1$.

The entire transfer function for the schematic will be equal to 1 as $H(\text{total}) = \text{multiplication of all transfer functions for the entire schematic}$. The 2 transfer function for the whole schematic were both equal to -1 meaning $H(\text{total}) = (-1)(-1) = 1$ so the input voltage should equal the output voltage of the whole schematic; the input voltage is labeled as V_{in} and is calculated using a voltage divider as shown before.

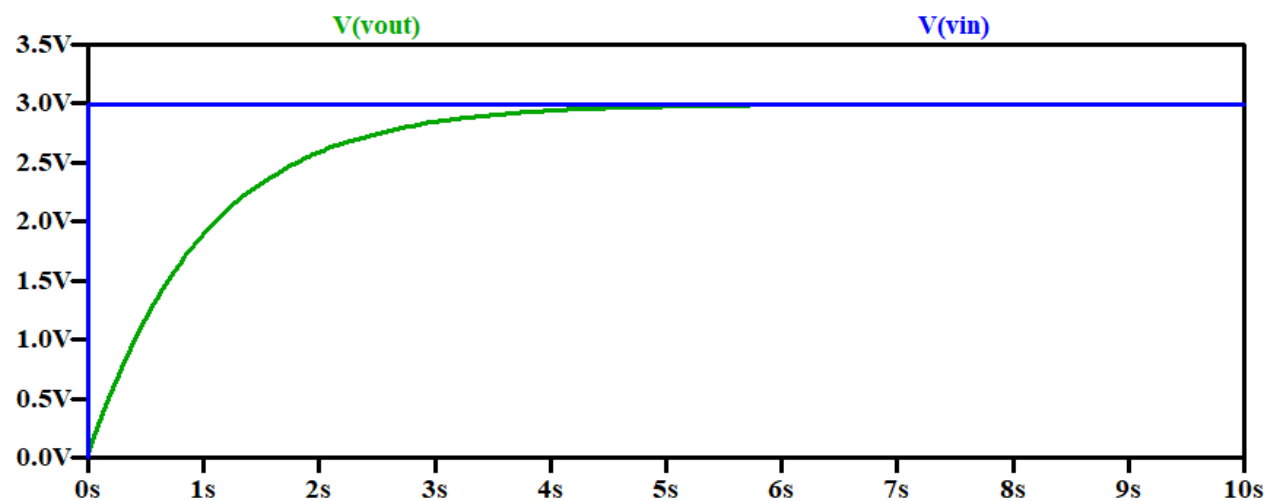
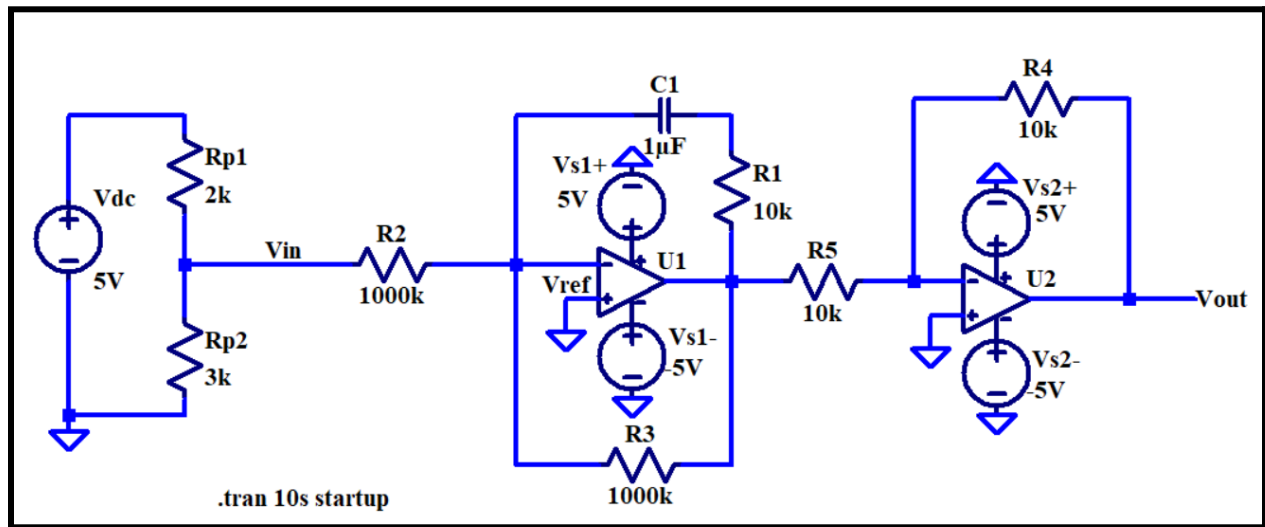
Simulation:

When V_{in} is 4.5V:



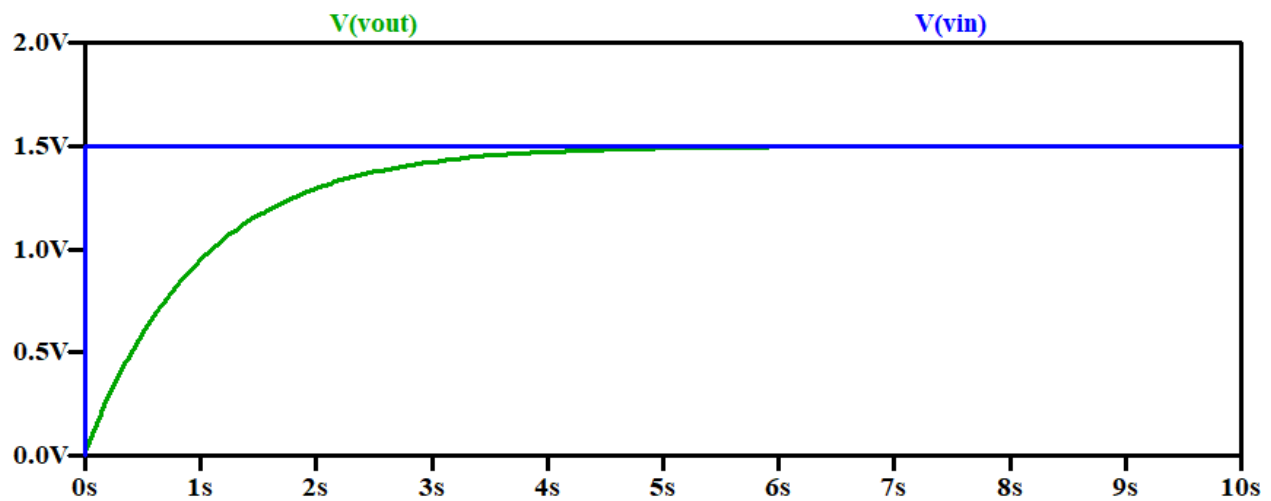
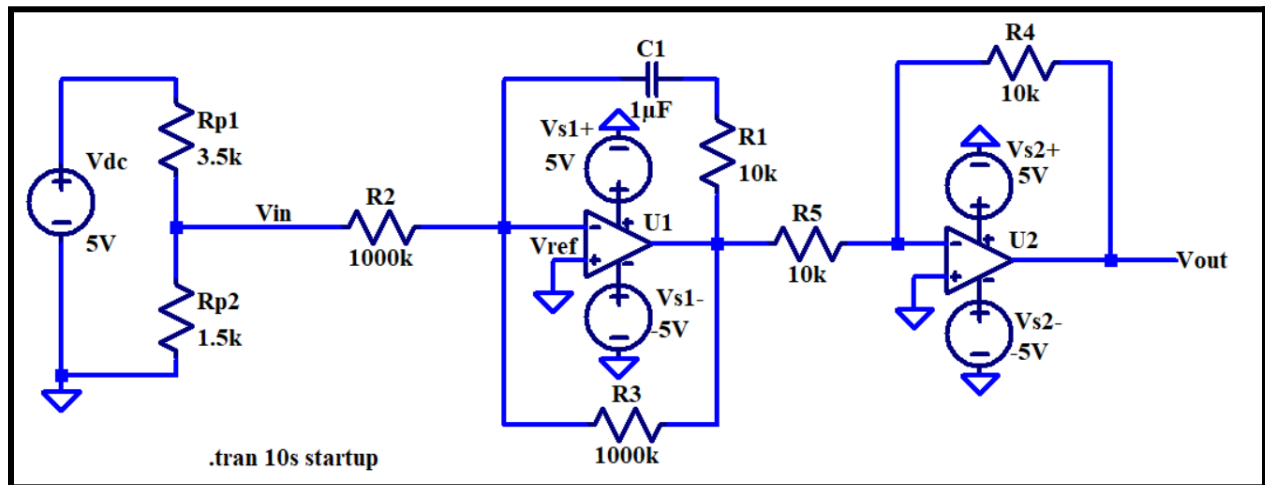
V_{out} is simulated to reach 4.5V

When V_{in} is 3V:



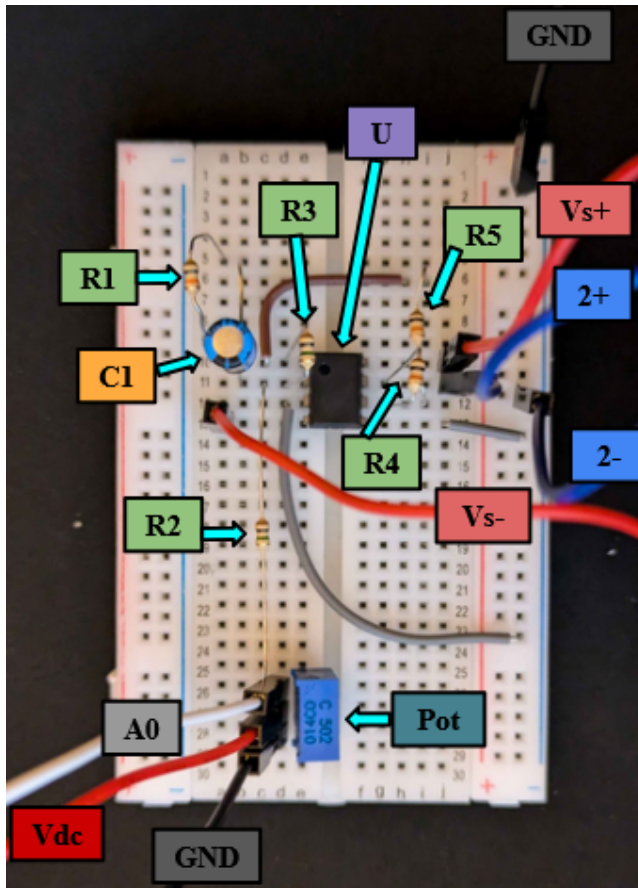
V_{out} is simulated to reach 3V

When V_{in} is 1.5V:



V_{out} is simulated to reach 1.5V

Measurement:



$$R1 = R4 = R5 = 10\text{k}\Omega$$

$$R2 = R3 = 1\text{M}\Omega$$

$$C1 = 1\mu\text{F}$$

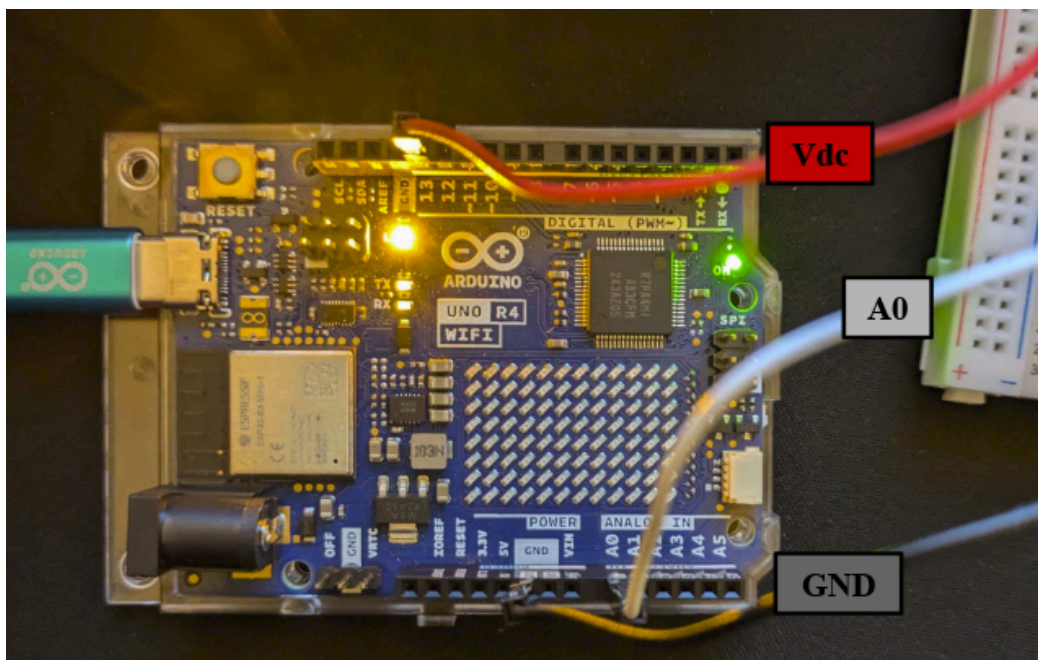
$$V_{s+/-} = +5/-5\text{V}$$

$$V_{dc} = +5\text{V}$$

U has 2 op-amps (TL072IP chip)

Pot is the 5k Ω potentiometer

Channel 2 measures Vout

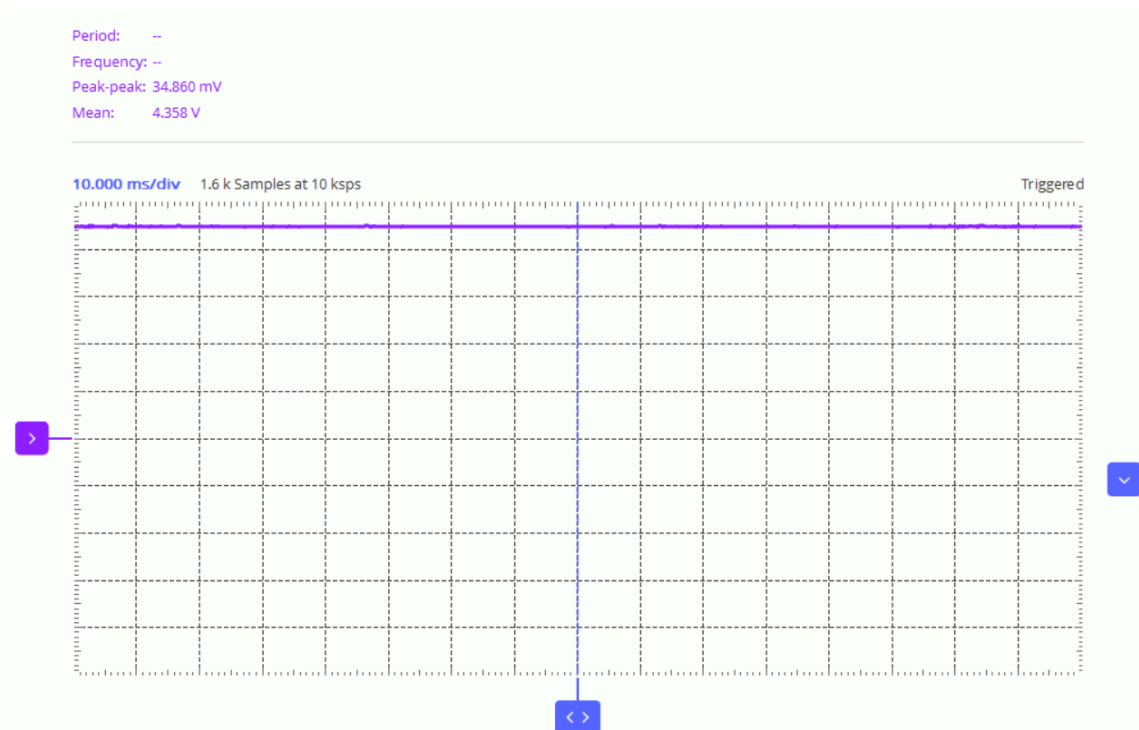


On MATLAB, to create a 5V dc source and have A0 read V_{in} from the potentiometer:

```
lab2.m  x  +
1      a = arduino();
2      writeDigitalPin(a, 'D13',1);
3      t=1;
4      while t>0
5          Vin = readVoltage(a, 'A0');
6          display(Vin);
7      end
8
```

When V_{in} is around 4.5V on MATLAB:

```
Vin =
4.4477
```

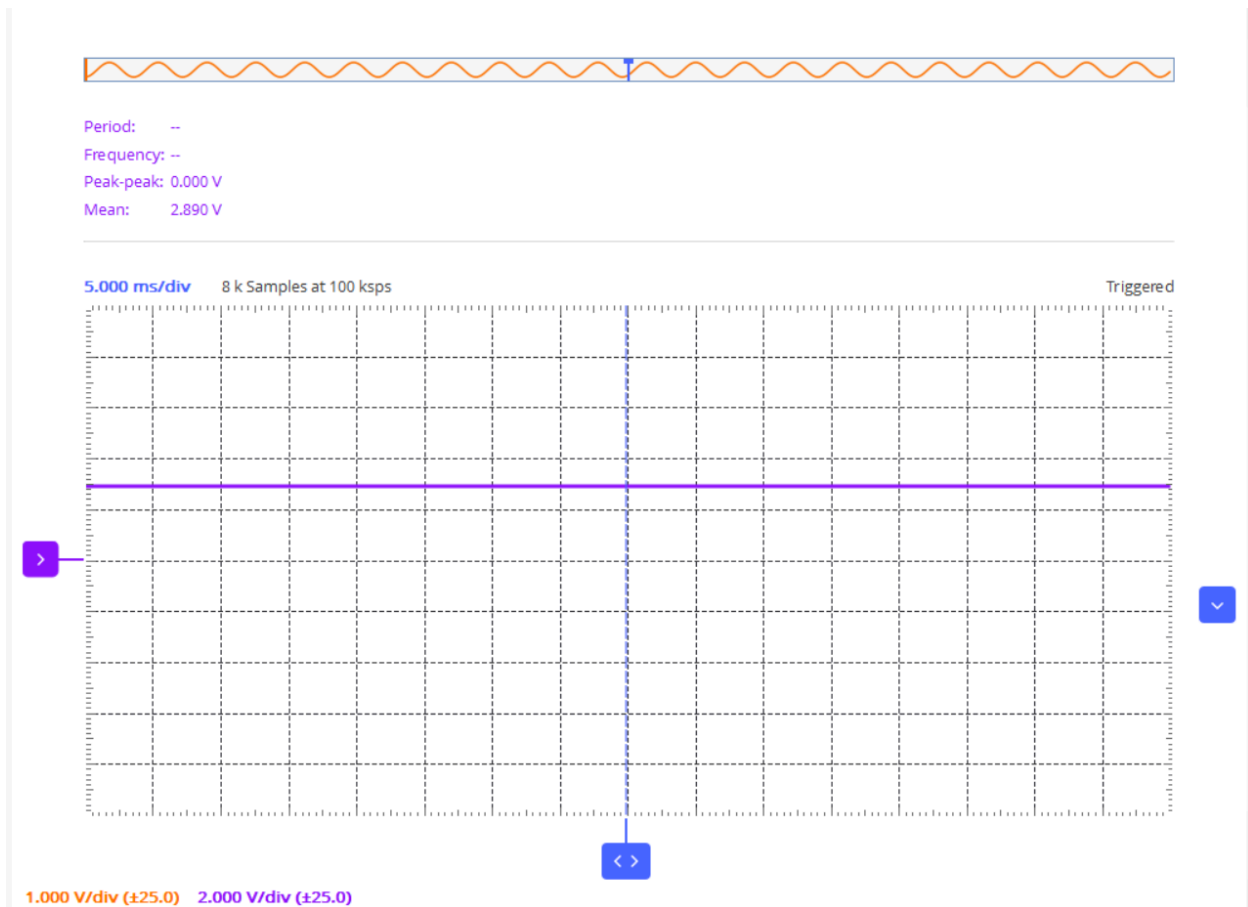


V_{out} is measured to be around 4.358V on scopy

When V_{in} is around 3V on MATLAB:

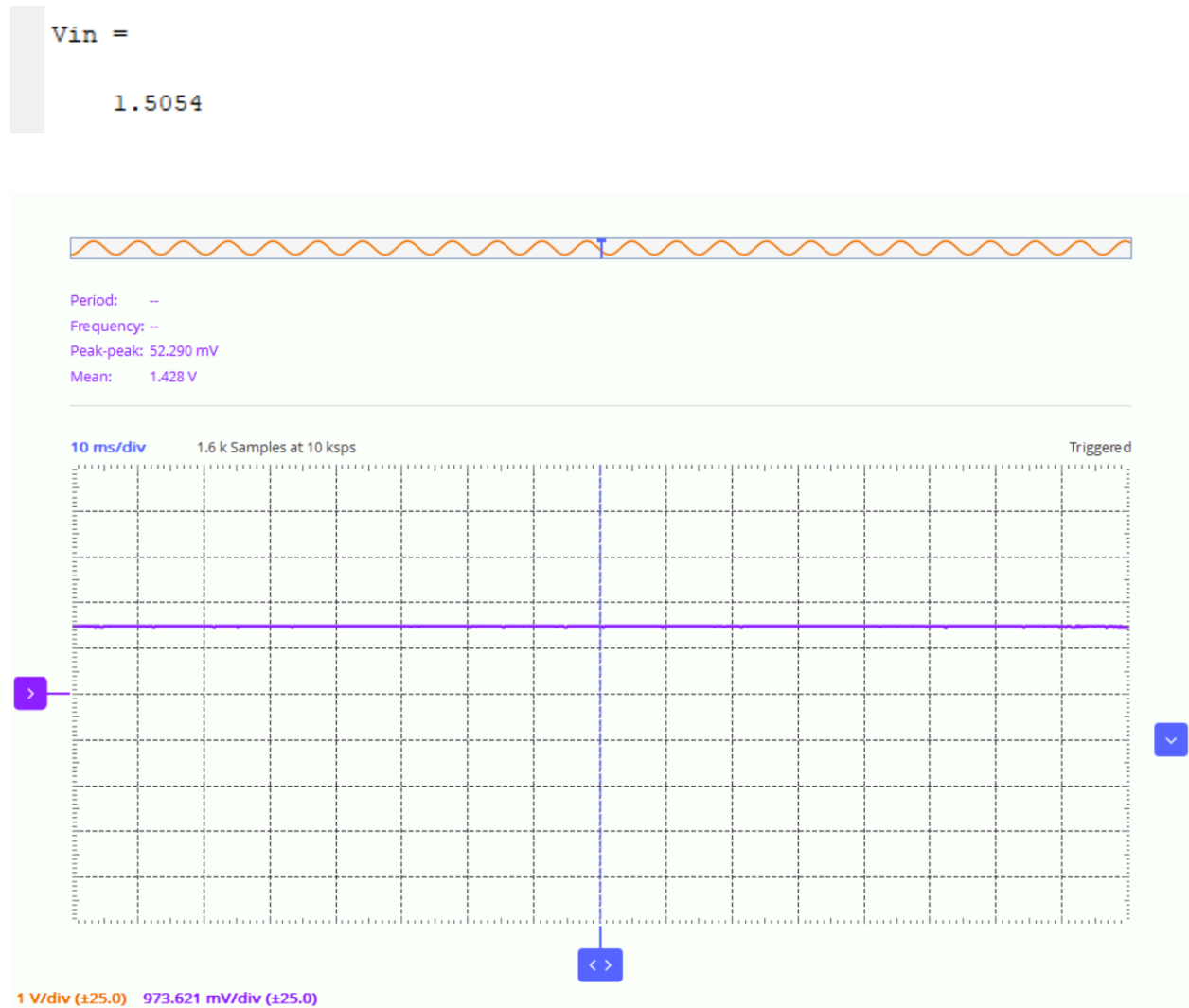
$V_{in} =$

3.0547



V_{out} is measured to be around 2.859 on scopy

When V_{in} is around 1.5V on MATLAB:



V_{out} is measured to be around 1.428V on scopy.

Discussion (and answer related questions in Alpha Lab):

Vref	Analysis	Simulation	Measurement	% Err
4.5V	4.5V	4.5V	4.345V	3.44%
3V	3V	3V	2.890V	4.70%
1.5V	1.5V	1.5V	1.428V	4.80%

The percent error for each is less than 5% which is reasonable since real life measurements don't come out to be as ideal as calculations or simulations due to circuit elements not being perfectly the value they are assumed to be. However, our error is low enough to conclude that the results of measurement are the values we expected to get and therefore allow us to prove that our PI controller design successfully produces a V_{out} that is about the value of V_{in} from an adjustable voltage source (via potentiometer).